

MLCK/MLCP regulates mammalian axon regeneration *via* the redistribution of the growth cone F-actin

Reviewed Preprint

v2 • December 13, 2024

Revised by authors

Reviewed Preprint

v1 • September 20, 2024

Saijilafu , Wei-Hua Wang, Jin-Jin Ma, Yin Yin, Yan-Xia Ma

Key Laboratory of Novel Targets and Drug Study for Neural Repair of Zhejiang Province, School of Medicine, Hangzhou City University, Hangzhou, China • Department of Orthopaedics, the First Affiliated Hospital, Orthopaedic Institute, Soochow University, Suzhou, China • Laboratory Animal Center, Suzhou Medical College of Soochow University, Suzhou, China

 https://en.wikipedia.org/wiki/Open_access
 Copyright information

eLife Assessment

Saijilafu et al. describe that MLCK and MLCP bidirectionally regulate NMII phosphorylation ultimately impinging on axonal growth during regeneration in the central and peripheral nervous systems. However, the evidence is in most cases **incomplete**, since some key controls are missing, some major claims are too broad to be supported by data and contrasting data on how MLCK and MLCP regulates NMII activity is not fully addressed or discussed. In sum, this knowledge is potentially **useful** for the field due to the relevance of identifying mechanisms that regulate axonal regeneration.

<https://doi.org/10.7554/eLife.101250.2.sa3>

Abstract

Axon regrowth is a key determinant of the restoration of the biological function of the nervous system after trauma. However, mature mammalian neurons have limited capacity for axon regeneration. We have previously demonstrated that neuronal axon growth both in the central and the peripheral nervous systems is markedly enhanced when non-muscle myosin II (NMII) is inhibited with blebbistatin. The activity of NMII is primarily regulated by MLCK and MLCP *via* the phosphorylation and dephosphorylation of its light chain, respectively; however, the functional roles of MLCK and MLCP in mammalian axonal regeneration remain unknown. In the present study, we provide strong evidence that the inhibition of MLCK activity significantly blocks axon regeneration in the mouse. Conversely, inhibition of MLCP promotes axon regrowth of both peripheral and central nervous system. Our findings further indicate that the MLCK/MLCP regulates axon regeneration via the reorganization of F-actin distribution in the growth cone, and this result suggests that direct regulation of the growth cone machinery is a potential strategy to promote axon regeneration.

Introduction

The coordinated regulation of growth cone cytoskeleton components is essential for axon growth. Additionally, cytoskeletal proteins in the growth cone are converging targets of extracellular inhibitory molecules. It is now recognized that extracellular inhibitory molecules, such as myelin-based inhibitors and chondroitin sulfate proteoglycans (CSPGs) in glial scars are a major hurdle for successful axon regeneration. These extracellular inhibitory molecules typically trigger the activation of the RhoA/ROCK pathway, leading to the phosphorylation and, consequently, the deactivation of cofilin, a protein that severs actin filaments and promotes their depolymerization. Cofilin deactivation results in the stabilization of the growth cone actin cytoskeleton protein, and, subsequently, growth cone collapse. Accordingly, in order to overcome those inhibitory substrates and promote axonal outgrowth, several recent studies have focused on manipulating growth cone cytoskeleton dynamics, and have obtained promising results^{1–4}. Here, we hypothesized that the appropriate spatiotemporal regulation of the local cytoskeletal machinery in the growth cone can provide an alternative and effective approach for enhancing axon regeneration.

We have previously shown that the inhibition of non-muscle myosin II (NMII) activity can promotes axonal growth over inhibitory substrates, such as CSPGs and myelin, in both the central and peripheral nervous systems (CNS and PNS, respectively)¹. Furthermore, this growth-promoting effect is achieved *via* the reorganization of microtubules (MTs) in the growth cone¹. Myosin proteins constitute a superfamily of ATP-dependent motor proteins known primarily for their role in muscle contraction. In the nervous system, however, myosin II usually drives actin retrograde flow in the growth cone, while the actin cytoskeleton powers directional growth cone motility^{5,6}. An increasing number of studies have shown that the actin cytoskeleton and its regulators also play a critical role in axon elongation and guidance. Myosin II is typically activated *via* the phosphorylation of the N-terminus of its light chain by myosin light chain kinase (MLCK) and deactivated through dephosphorylation by myosin light chain phosphatase (MLCP). An early study reported that the expression of MLCK, a calcium/calmodulin-dependent kinase, is increased during axonal outgrowth in goldfish retinal ganglion cells⁷. The overexpression of constitutively active MLCK in *Drosophila* CNS neurons increases myosin II activity and leads to impaired axonal outgrowth⁸. Zhang *et al.* demonstrated that MLCK inhibition attenuates 5-HT-dependent neurite outgrowth in bag cell neurons of the mollusk *Aplysia*⁹. Additionally, the activation of MLCP/MLC signaling in PC12 cells was also shown to promote neurite outgrowth¹⁰. Together, these reports suggested that the MLCK/MLCP regulates axon growth in non-mammalian organisms. However, to our knowledge, the functional roles of MLCK and MLCP during mammalian axon regeneration remain unknown.

In the present study, we found that sciatic nerve injury elevated MLCK protein expression, leading to increased levels of MLC phosphorylation in dorsal root ganglion (DRG) neurons. Furthermore, the pharmacological inhibition or genetic silencing of MLCK blocked axon regeneration in the PNS and CNS both *in vitro* and *in vivo*, whereas the pharmacological inhibition or transcriptional silencing of MLCP dramatically promotes axon regeneration. We further found that the MLCK/MLCP activity also affects F-actin redistribution in the growth cone. Taken together, our results indicated that MLCK and MLCP may regulate axon regeneration *via* the redistribution of F-actin in the growth cone, and the direct manipulation of cytoskeleton proteins in the growth cone may be a promising strategy for enhancing mammalian axon regeneration.

Materials and methods

Animals and reagents

Adult ICR (8–10 weeks old, 20–30 g) were used. BDA (D1956) and CTB labeled with Alexa Fluor 488 (CTB-488, C34775) were obtained from Life Technologies. ML-7 hydrochloride (110448-33-4) was purchased from Santa Cruz Biotechnology. PDBu (P1269) was acquired from Sigma-Aldrich. Blebbistatin (S7099) was obtained from Selleck Chemicals. Antibodies against β III-tubulin (TUJ1, T8578), MLCK (SAB1300116), and GFAP (G3893) were purchased from Sigma-Aldrich. Antibodies against β -actin (#4970) and p-MLC (#3675) were acquired from Cell Signaling Technology. Actin-stain 555 phalloidin (Cat[#] PHDH1) was purchased from Cytoskeleton, Inc. siRNAs designed to target mouse MLCK (siMLCK) (5'-GGCAAATACACCTGTGAAG-3', 5'-TGGTCAAAGAAGGGCAGAT-3', 5'-AGCCAAAGGGAGTCAACAT-3', and 5'-TCACGACGGGAATGAGATT-3') and mouse MYPT1 (siMYPT1) (5'-CTGTGGATATCTCGATATTGC-3') were obtained from GenePharma (Shanghai, China).

Cell culture

Adult DRG neurons were dissociated from 8 to 10-week-old ICR mice and cultured in Minimal Essential Medium (MEM) containing 5% (v/v) fetal bovine serum (FBS), 5-fluoro-2-deoxyuridine/uridine (20 μ M), and penicillin/streptomycin. Embryonic cortical neurons (from E14.5 embryos) and hippocampal neurons (from E18.5 embryos) were cultured in neurobasal medium supplemented with penicillin/streptomycin, GlutaMAX, and B27 supplement. To investigate axonal growth on inhibitory substrates, glass coverslips were first coated with poly-D-lysine (100 μ g/mL) for 2 h, and then coated again with myelin or 10 μ g/mL CSPGs for 2 h, as previously described^{12,11}.

For the cell replating experiment, DRG neurons were cultured for 3 days, resuspended in culture medium, replated on new coverslips, and finally re-cultured for a further 18 h to allow new axons to grow, as previously described^{12,13}. Briefly, in treatments denoted as “before replating”, neurons were first grown in the presence of 10 μ M ML-7 during the initial 3-day culture period, and, after washing out the ML-7, they were replated and grown in the absence of drug(s) for 18 h. Treatments denoted as “after replating” refer to procedures in which neurons were first grown in the absence of a drug or drugs during the initial 3-day culture period, after which 10 μ M ML-7 was added immediately after replating, followed by 18 h of culture in the presence of drug(s).

Sciatic nerve axotomy

The sciatic nerve was transected at the sciatic notch using ophthalmic scissors. After surgery, the wound was closed, and the mice were allowed to recover. The mice were then euthanized and L4–L5 DRGs were isolated for cell culture, quantitative reverse transcription-PCR (qRT-PCR), and western blotting.

siRNA transfection

Cells were transfected with siRNA by electroporation in accordance with the manufacturer's instructions. Briefly, dissociated neurons were suspended in 80- μ L solutions of Amaxa electroporation buffer containing siRNAs (2–3 nmol) and/or an EGFP-expressing plasmid (5 mg). The cells were then transferred to a 2-mm cuvette and electroporated using an Amaxa Nucleofector apparatus. After electroporation, the cells were immediately transferred to 500 μ L of prewarmed culture medium and cultured on glass coverslips coated with 100 mg/mL poly-D-lysine. After 4 h, when the neurons had fully attached to the coverslips, the remaining electroporation buffer was discarded, and fresh culture medium (500 mL) was added. We used the pCMV-EGFP-N3 as control, and the pCMV-EGFP-N3 plasmid was from Clontech, Inc.

qRT-PCR and western blotting

Total RNA was isolated using Trizol Reagent and assessed for integrity and concentration in a NanoDrop 1000 spectrophotometer. Each sample was then reverse transcribed using Maxima H Minus Reverse Transcriptase. Real-time qPCR was performed using SYBR Premix ExTaq II in a CFX96 Real-Time qPCR Detection System (Bio-Rad). Glyceraldehyde 3-phosphate dehydrogenase (*Gapdh*) was used as the internal control. The sequences of the forward and reverse primers were MLCK: 5'-AGAAGTCAAGGAGGTAAAGAATGATGT-3' and 5'-CGGGTCGCTTTTCATTGC-3', respectively; and GAPDH: 5'-AGGTCGGTGTGAACGGATTTG-3' and 5'-TGTAGACCATGTAGTTGAGGTCA-3', respectively.

For western blotting, samples were homogenized in RIPA buffer. After boiling for 5 min, equal amounts of extracted protein (20 µg/sample) were separated using 10% sodium dodecyl sulfate–polyacrylamide gel electrophoresis, and electro-transferred to polyvinylidene fluoride membranes (Immobilon-P; Millipore). After blocking with 0.01 M PBS containing 5% (w/v) skimmed milk overnight, the membranes were incubated first with primary antibodies and then with the appropriate secondary peroxidase-conjugated antibodies. Protein bands were developed using ECL Prime Western Blotting Detection Reagent and subsequently quantified using Image Lab software. An anti-β-actin antibody was used as a loading control.

Immunohistostaining

Mice were deeply anesthetized and transcardiacally perfused with 4% paraformaldehyde (PFA). L4–L5 DRG tissues were isolated, further fixed in 4% PFA, cryosectioned into 12-µm slices, and blocked with 5% FBS containing 0.3% Triton X-100. The sections were then incubated with the indicated primary antibody (monoclonal anti-βIII-tubulin antibody and/or polyclonal anti-MLCK antibody). Immunoreactivity was visualized using the appropriate fluorescently labeled secondary antibody. Alternatively, cultured DRG neurons were fixed in 4% PFA, blocked with 2% BSA and 0.1% Triton X-100, and then stained as described above. To visualize F-actin in the growth cones, *in vitro*-cultured neurons were stained with actin-stain 555 phalloidin following the manufacturer's instructions.

Axonal growth and growth cone quantification

All images were acquired at 1388 × 1040-pixel resolution using a Zeiss fluorescence microscope with AxioVision 4.7 software (Carl Zeiss Micro Imaging, Inc.) equipped with a 10× or a 20× objective. To quantify axon length, neurons with axonal processes exceedingly twice the diameter of their cell body were selected, and the longest axon from 100 neurons for each condition was measured using the curve tool in AxioVision 4.7. Meanwhile, growth cone size and F-actin area were measured using the outline tool in AxioVision 4.7. The size of the growth cone was ascertained from the neck of the microtubule to the edge of the axon tip and the F-actin area was determined from the hillock to the most distant edge of the growth cone.

In vivo electroporation of DRGs and sciatic nerve crush

In vivo electroporation of adult mouse DRGs was performed as previously described¹⁴. Briefly, each mouse was anesthetized *via* an intraperitoneal injection of a ketamine (100 mg/kg) and xylazine (10 mg/kg) solution. The transverse process of L4 and L5 was carefully removed to expose the DRG tissue. A 1 µl solution containing indicated siRNAs together with the plasmid encoding EGFP (pCMV-EGFP-N3) was then microinjected into the L4–L5 DRG tissue using a Picospritzer II Microcellular Injection Unit followed by electroporation using a BTX ECM830 Electro Square Porator (five 15-ms pulses, 35 V, 900-ms interval). Following this, the muscle and skin were carefully sutured using 5-0 nylon sutures. The mouse was placed on a heated blanket (35°C) until it had completely recovered from anesthesia. Two days after electroporation, the sciatic nerves were crushed with forceps #5 and the crush sites were marked with an 11-0 nylon epineuria suture.

After 3 more days, the mice were euthanized by perfusion of 4% PFA, and the whole sciatic nerves were dissected out and postfixed overnight in 4% PFA at 4°C. Prior to the whole-mount flattening process, it was essential to verify that the location of the epineural suture corresponded with the site of injury. Only experiments where the crush site was distinctly recognizable were considered for inclusion in the subsequent analysis. In order to quantify axon regeneration, all discernible axons labeled with EGFP within the sciatic nerve were manually traced and measured from the crush site to the distal tip.

Optic nerve injury

The optic nerves of adult C57BL/6 mice were carefully exposed and crushed with a jeweler's forceps #5 just behind the eyeball for 1 s. A small piece of gelatin sponge fully soaked with PBS or 50 μ M PDBu was placed at the lesion site; alternatively, 2 μ L of PDBu (50 μ M) was intravitreally injected into the vitreous body. Simultaneously, 2 μ L of CTB-488 was injected into the vitreous body of each animal after the optic nerve had been crushed. All microinjections were performed using the Picospritzer II Microcellular Injection Unit (Parker Ins.; pressure: 15 psi, duration: 6 ms). After 3 days, the optic nerves were dissected out, fixed in 4% PFA overnight at 4°C, and cut into a series of 12- μ m longitudinal sections. The numbers of CTB-labeled axon fibers were counted at distances of 50 and 100 μ m from the injury site. The length of the longest CTB-labeled axon fiber of each nerve was also measured.

Spinal cord crush injury and PDBu injection

The T8 spinal cord was exposed under a microscope, the whole spinal cord was crushed for 2 s with a modified jeweler's forceps #5 as previously described¹⁵, and the muscle and skin were separately sutured. A 100- μ L volume of PDBu (50 μ M) or DMSO was administered every 4 days by intrathecal injection. After 14 days, to trace and visualize the corticospinal tract axon, 1.6 μ L of 10% BDA was injected into 4 sites of the sensorimotor cortex (1.0 mm lateral; 0.5 mm deep into the cortex; 1.0, 0.5, -0.5, and -1.0 posterior to bregma) with a 5 μ L Hamilton syringe¹⁶. Fourteen days after BDA injection, mice were transcardially perfused with 4% PFA. The whole spinal cord was isolated from each mouse and post-fixed in 4% PFA for 6 h at 4°C. After dehydration using an increasing gradient of sucrose concentrations, the spinal cord was cut into 25- μ m-thick sagittal sections, which were then stained with Cy3-streptavidin for 3 h following the manufacturer's protocol. The lengths of regenerating axons were determined by measuring the distance from the lesion site to the tip of the BDA-labeled axon tip. The final axon length was the average calculated from 5 sections for each spinal cord. To quantify the retraction bulb, both maximum width of the enlarged distal tip of the axon and the width of its immediately adjacent axon shaft was measured. Then, the ratio of these two widths was then calculated. An axonal tip was considered as a retraction bulb if its tip/shaft ratio exceeded 4. Averages number of retraction bulb were calculated from 3 sections in every mouse for each group. Five mice were used for each condition.

BBB locomotor scale score assessments

BBB score assessments were performed as previously described¹⁷. Briefly, mice were individually placed in an open, quiet space (45 $\times$ 30 cm) for 5 min while two individuals blinded to the experiments scored the hindlimb motor function of each animal. Assessments were made on day -1 (the day before injury) and on days 1, 3, 7, 14, 21, 28, 35, and 42 post-injuries for both the DMSO and PDBu treatment groups.

Results

The MLCK expression increased in adult sensory neurons following peripheral axotomy

Following injury, the expression of axon regeneration-related genes is usually upregulated. Thus, we first measured MLCK protein levels in DRG neurons following sciatic nerve transection. Compared with naive neurons, MLCK protein expression was significantly increased in DRG neurons 3 days after sciatic nerve transection (**Figure 1A**, B). The mRNA levels of MLCK in DRG neurons were also increased on days 1, 3, and 7 after axotomy, with the highest expression being observed on day 3 (**Figure 1C**). Given that MLCK phosphorylates MLC, we subsequently assessed the level of MLC phosphorylation and found that it was higher in DRG neurons following nerve injury than in control DRG neurons, likely due to the increase in MLCK expression (**Figure 1A**, D). Immunohistochemical staining further confirmed that both MLCK expression and the levels of phosphorylated MLC (p-MLC) increased in DRG neurons following sciatic nerve axotomy (**Figure 1E**, F). Taken together, these results indicated that MLCK is activated in DRG sensory neurons following peripheral axotomy, which leads to an increase in MLC phosphorylation.

The inhibition of MLCK impaired axon regeneration in peripheral sensory DRG neurons

Next, to explore the functional role of MLCK during peripheral axon regeneration, ML-7 (10 μ M), a specific pharmacological inhibitor of MLCK, was added to DRG neuron cultures for 3 days. The results showed that ML-7 administration led to a significant reduction in MLC phosphorylation levels (**Figure 2A**, B, C) and impaired axonal growth in sensory neurons (**Figure 2D**, E), and these effects were dose-dependent (Supplementary Figure S1). The proportion of neurons with axons also decreased after ML-7 treatment (**Figure 2E**). Furthermore, the transfection of a small interfering RNA (siRNA) targeting MLCK (siMLCK) into dissociated DRG neurons significantly inhibited MLCK expression and reduced p-MLC levels (**Figure 2F**, G), while concomitantly blocking axon growth in adult DRG sensory neurons (**Figure 2H**, I). Next, using our previously described *in vivo* DRG electroporation technique¹⁴, we further explored the functional role of MLCK in sciatic nerve axon regeneration *in vivo*. We co-transfected siMLCK and an EGFP-expressing plasmid into lumbar vertebrae 4 and 5 (L4 and L5) DRGs of adult mice by electroporation. After 2 days, mice were subjected to sciatic nerve crush injury, and, after another 3 days, the lengths of regenerating axons were measured in whole-mount sciatic nerves. Our data showed that the mean length of *in vivo* regenerating axons was significantly reduced in siMLCK-treated mice compared with that in control animals (**Figure 2J**, K). These findings suggested that MLCK activity is required for mammalian peripheral axon regeneration both *in vitro* and *in vivo*.

The inhibition of MLCP promoted peripheral axon growth over inhibitory substrates

It is well known that the opposite roles of MLCK and MLCP in regulating the MLC phosphorylation status. Consequently, we also investigated the functional role of MLCP during axon regeneration. It has been reported that phorbol 12,13-dibutyrate (PDBu) can block MLCP activity, which leads to an increase in MLC phosphorylation¹⁸. Here, the results demonstrated that PDBu treatment (20 μ M) significantly increased the p-MLC level in cultured DRG neurons (Supplementary Figure S2A, B). Importantly, we found that this occurred in parallel with greatly enhanced axonal growth in DRG neurons cultured on permissive substrates, such as poly-D-lysine (Supplementary Figure S2C, D). A key reason why CNS neurons cannot spontaneously regenerate axons after injury is due to the inhibitory local microenvironment. Myelin and CSPGs are well-characterized CNS-based inhibitory substrates that block axon regeneration. Therefore, we next examined whether inhibition of MLCP activity with PDBu promoted axon regeneration in the presence of these potential CNS-based inhibitory substrates. When dissociated adult DRG neurons were cultured on myelin extracts or

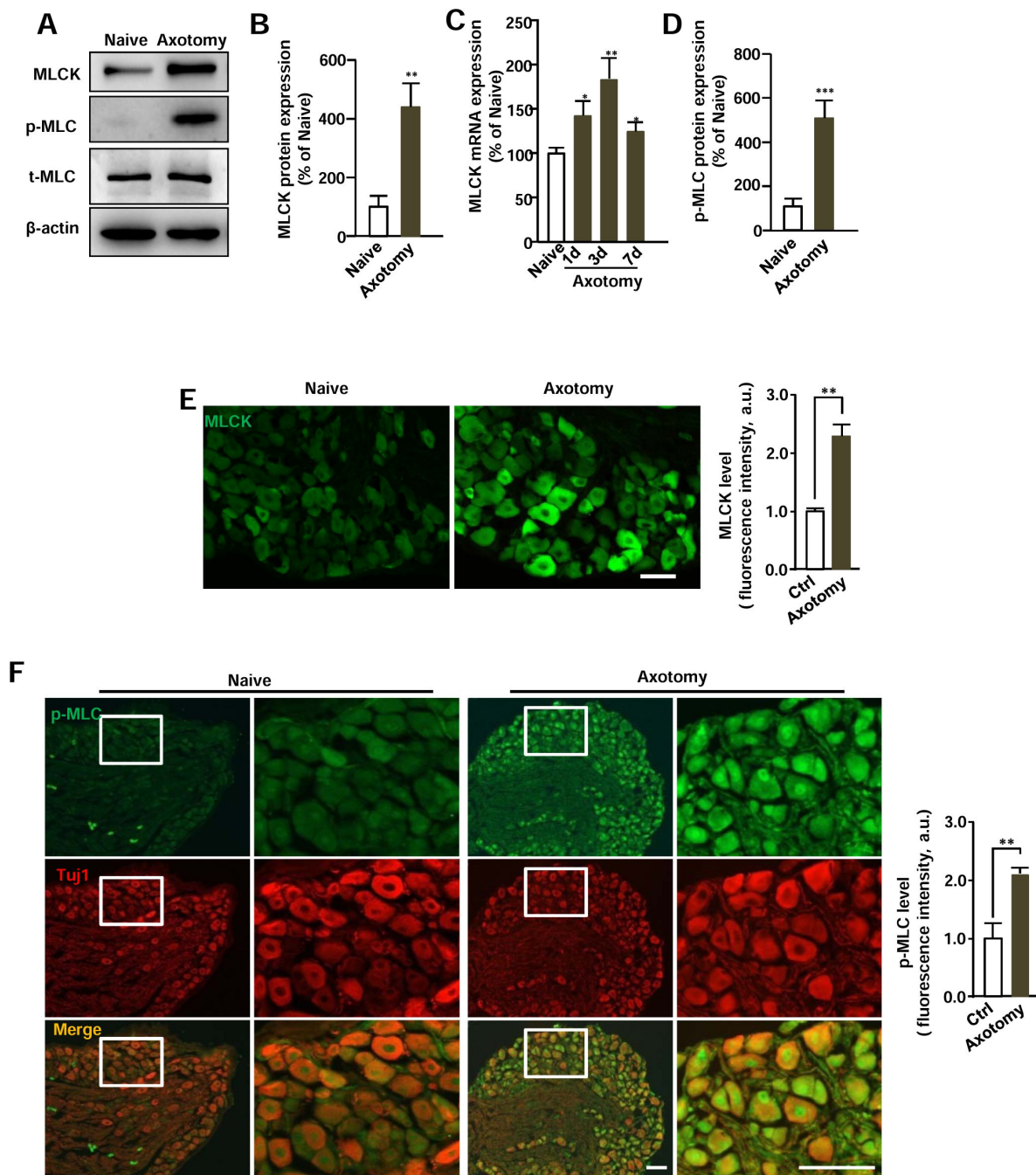


Figure 1.

MLCK and phosphor-MLC expression increased in sensory neurons after peripheral axotomy.

(A-D) Sciatic nerve axotomy was performed 3 days before and L4-L5 DRG tissues were collected for Western blotting or qPCR to examine the expression level of MLCK and phosphor-MLC (p-MLC). Representative images (A) and quantification of Western blots (B, D) demonstrate that expression levels of MLCK and p-MLC increase significantly in adult DRGs following sciatic nerve transection. (N=3, ** $p < 0.01$; *** $p < 0.001$).

(A) (C) L4-L5 DRGs were collected for qRT-PCR, 1, 3 and 7 days after axotomy, respectively. The results reveal that mRNA expression of MLCK in the DRG neurons also increased after axotomy after 1, 3 and 7 days, although the difference at 3 days was the most significant. (N=3, * $p < 0.05$; ** $p < 0.01$).

(E-F) Sections of L4-L5 DRGs from naive and sciatic nerve axotomized mice were stained with antibodies against MLCK, Tuj-1 or p-MLC. Immunohistochemistry images indicate that MLCK (E) and p-MLC (F) expression were elevated in the sensory neurons following sciatic nerve axotomy. Scale bar: 100 μ m.

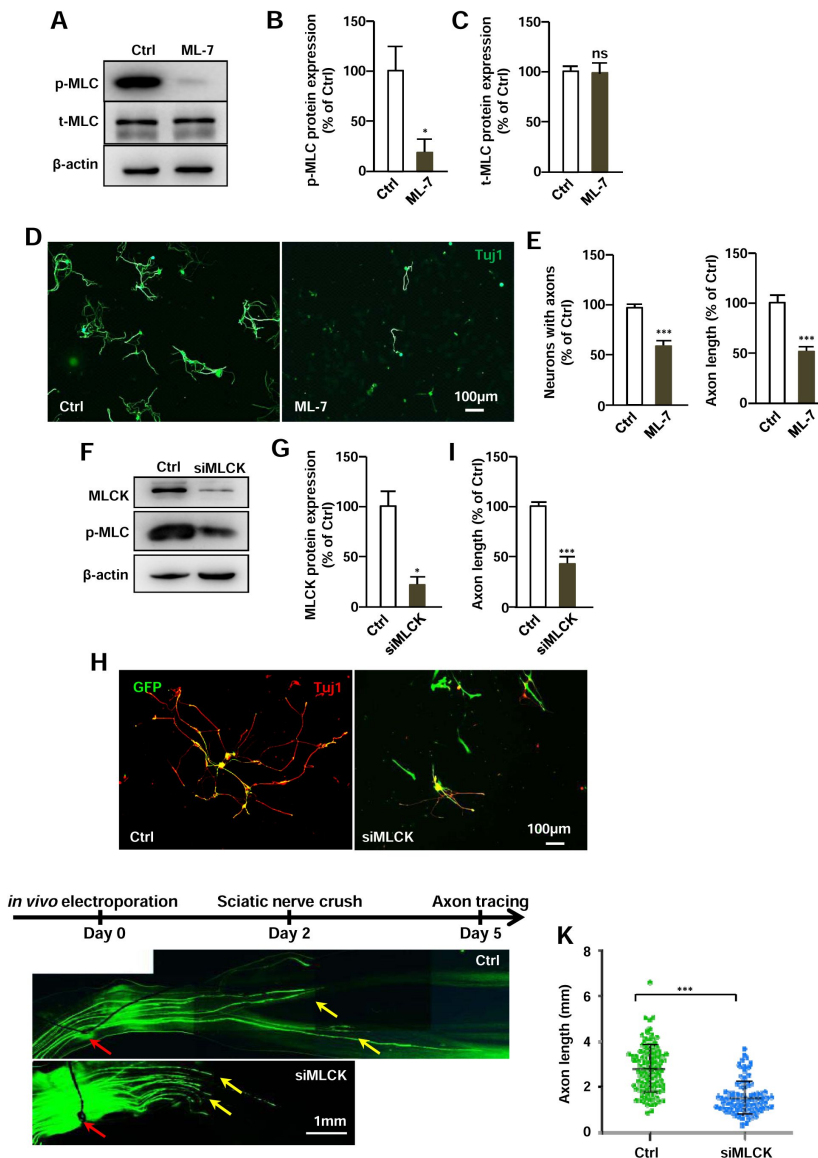


Figure 2.

Suppression of MLCK activity inhibits axon growth in adult DRG neurons.

(A-E) DRG neurons from adult mice were cultured and treated with DMSO or 10.0 μ M ML-7. After 3 days, neurons were lysed for Western blotting or staining with Tuj1 antibody. Western blot images (A) and quantification of p-MLC (B) indicate that ML-7 treatment successfully inhibits MLCK activity. Quantification of axon length indicated that inhibition of MLCK activity with ML-7 dramatically blocks adult sensory neuronal axon growth *in vitro* (D, E), and reduces the numbers of neurons with axons (E). Scale bar: 100 μ m (N=3, * p < 0.05; *** p < 0.001).

(F-I) Adult DRG neurons were isolated and electroporated with scramble siRNA (Ctrl) or MLCK siRNA (siMLCK) mixed with GFP. Tuj1 (Red) antibody was used to stain neurons 3 days after transfection. A representative image (F) and quantification of Western blots for MLCK (G) indicate that siMLCK significantly suppressed MLCK and p-MLC expression (N=3, * p < 0.05). Quantification of mean axon length (H, I) indicates that transfection of siMLCK inhibits adult DRG neuronal axon growth. Scale bar: 100 μ m. (N=3, *** p < 0.001).

(J, K) L4/L5 DRGs of adult mice were electroporated with scramble siRNA (Ctrl) or siMLCK combined with GFP, respectively. Two days later, sciatic nerves were crushed and then regenerating axons traced after a further three days. Representative images (J) and measurement of regenerating axonal length (K) indicated that decreased MLCK expression with siRNA inhibits sensory axon regeneration *in vivo*. Crush sites are marked using red arrow. Regenerating axon tips are marked with yellow arrow (N=5, * p < 0.05). Scale bar: 1mm.

CSPGs, sensory neuronal axon growth was significantly suppressed (Supplementary Figure S2C, D). Interestingly, however, when PDBu was added to the culture medium, axon growth was significantly enhanced on either substrate (Supplementary Figure S2C, D). This result indicated that PDBu treatment relieved the inhibitory effect of myelin and CSPGs on axonal growth in DRG neurons. MLCP is a phosphatase complex comprising PP1 δ , MYPT1, and M20¹⁹. In this study, a siRNA specifically targeting MYPT1 (siMYPY1) was found to markedly increase MLC phosphorylation in cultured DRG neurons (Figure 3A, B). In line with the results obtained with the pharmacological inhibitor, siMYPT1 also significantly stimulated axonal growth in DRG sensory neurons cultured on myelin or CSPGs (Figure 3C, D). Furthermore, the transfection of siMYPT1 into L4–L5 DRGs of adult mice by electroporation also significantly promoted sciatic nerve axon regeneration *in vivo* (Figure 3E, F). Combined, these data indicated that the inhibition of MLCP activity enables mammalian axon regeneration *in vitro* and *in vivo*.

MLCK/MLCP activity regulated axon growth in the embryonic CNS *in vitro*

Compared with neurons of the PNS, those of the CNS have intrinsically poor axonal growth capability, which also helps explain why CNS neurons cannot regenerate axons spontaneously after injury. Therefore, we next examined the role of MLCK/MLCP on CNS axon regeneration using *in vitro* cultured cortical and hippocampal neurons from embryonic day (E)14.5 and E18.5 mice, respectively. To block MLCK activity, the neurons were treated with 10 μ M ML-7 for 3 days. The results showed that the MLC phosphorylation levels were markedly inhibited with ML-7 treatment (Figure 4A); simultaneously, the mean axon length of ML-7-treated E14.5 cortical neurons was significantly reduced compared with that of the control neurons (Figure 4C, D). Furthermore, consistent with the pharmacological results, the siRNA-mediated suppression of MLCK expression in E14.5 cortical neurons (Figure 4B) significantly inhibited their axonal growth (Figure 4E, F). Conversely, the blockade of MLCP activity with PDBu (20 μ M; Figure 4A) notably augmented cortical neuronal axonal growth (Figure 4C, D). The siRNA-mediated transcriptional silencing of MYPT1 expression also upregulated p-MLC levels in cortical neurons (Figure 4B), and, accordingly, also enhanced axonal growth (Figure 4E, F). Similar results were obtained with E18.5 hippocampal neurons (Figure 4G–J). Overall, these findings demonstrated that MLCK/MLCP activity also regulates axon growth in the developing CNS.

The local inhibition of MLCP activity at an injury site promoted optic nerve regeneration

Given the promising axon growth-promoting effects observed *in vitro*, we wondered whether inhibiting MLCP activity could stimulate *in vivo* CNS axon regeneration. The optic nerve forms part of the CNS and, like other neurons of the CNS, also has weak intrinsic axon-regenerating ability (Supplementary Figure S3). To test this, the right optic nerve of adult mice was crushed, and PBS or PDBu (50 μ M) was applied locally to the lesion site using a gelatin sponge; alternatively, 2 μ L of PDBu (50 μ M) was microinjected into the vitreous body of another mice. Subsequently, 2 μ L of Alexa Fluor 488-conjugated cholera toxin subunit B (CTB) was injected into the vitreous body to anterogradely label regenerating axons. After 3 days, the whole optic nerve was harvested, and CTB-labeled axon regeneration was assessed. The local administration of PDBu using a gelatin sponge resulted in numerous CTB-labeled axon fibers growing over the injury site, the longest being approximately 400 μ m long (Figure 5A–C). In contrast, local PBS administration at the injury site or intravitreal PDBu injection did not significantly enhance axon regeneration beyond the injury site (Figure 5A–C). This observation suggests that the only treatment employed in the injury site (the inhibition of MLCP activity within the growth cone) effectively promote axonal growth.

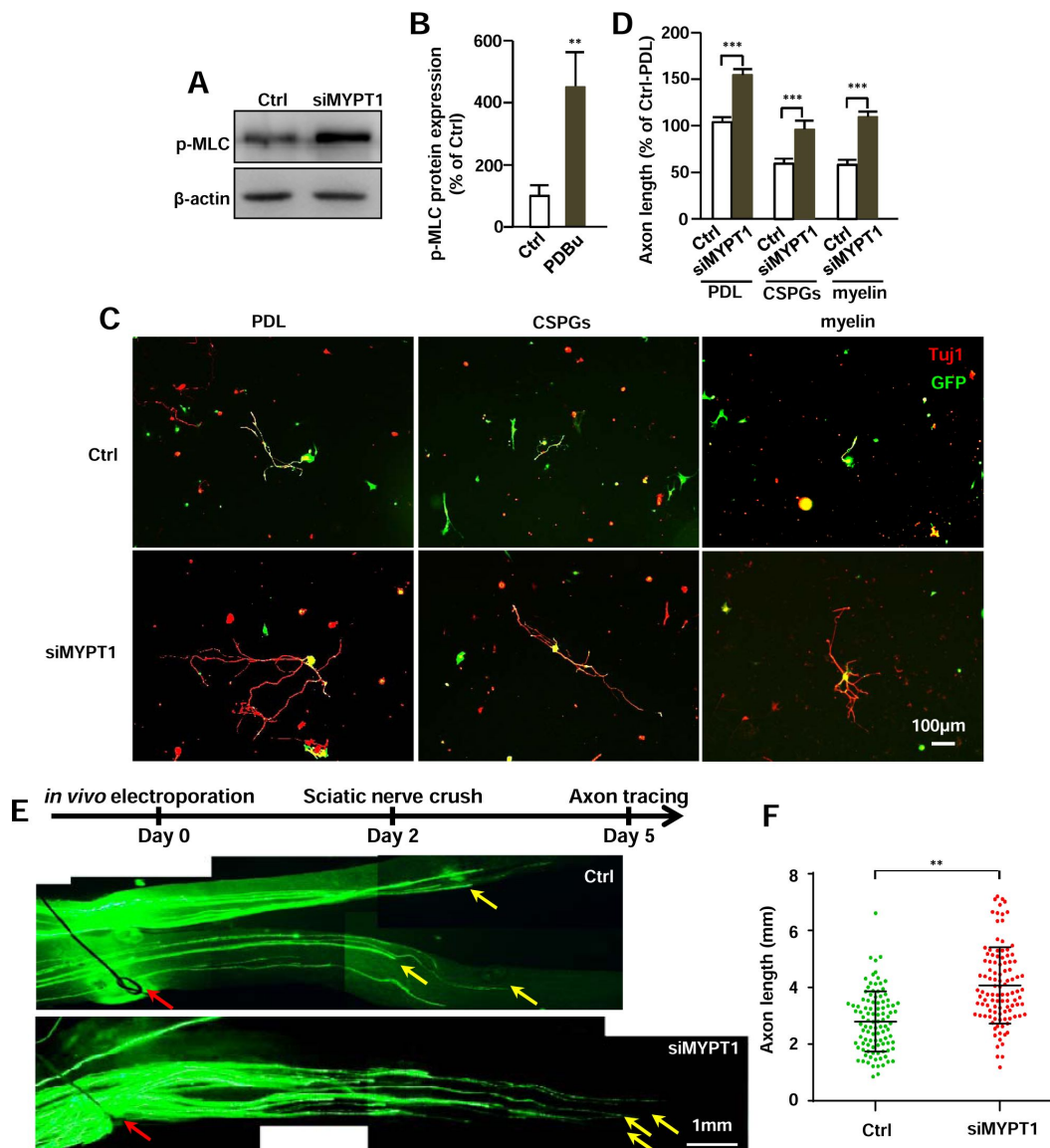


Figure 3.

Knockdown of MLCP protein promotes axon growth from adult peripheral sensory neurons.

(A-D) Adult DRG neurons were isolated and electroporated with scramble siRNA (Ctrl) or MYPT1 siRNA (siMYPT1) mixed with GFP. A representative Western blot image (A) and quantification of protein band (B) indicate that transfection of siMYPT1 in cultured DRG neurons significantly increases p-MLC expression ($N=3$, $**p < 0.01$). Quantification of axon length (C, D) indicated siMYPT1 transfection in the adult sensory neurons promotes its axon growth on the permissive substrate PDL or inhibitory substrates, such as CSPGs and myelin. ($N=3$, $***p < 0.001$).

(E, F) L4/L5 DRGs of adult mice were electroporated with scramble siRNA (Ctrl) or siMYPT1, respectively. Two days later, sciatic nerves were crushed and then regenerated axons traced after a further three days. Representative images (E) and measurement of regenerating axonal length (F) indicated that siMYPT1 promotes sensory axon regeneration *in vivo*. Crush sites are marked using red arrow. Regenerating axon tips are marked with yellow arrow ($N=5$, $*p < 0.05$). Scale bar: 1mm.

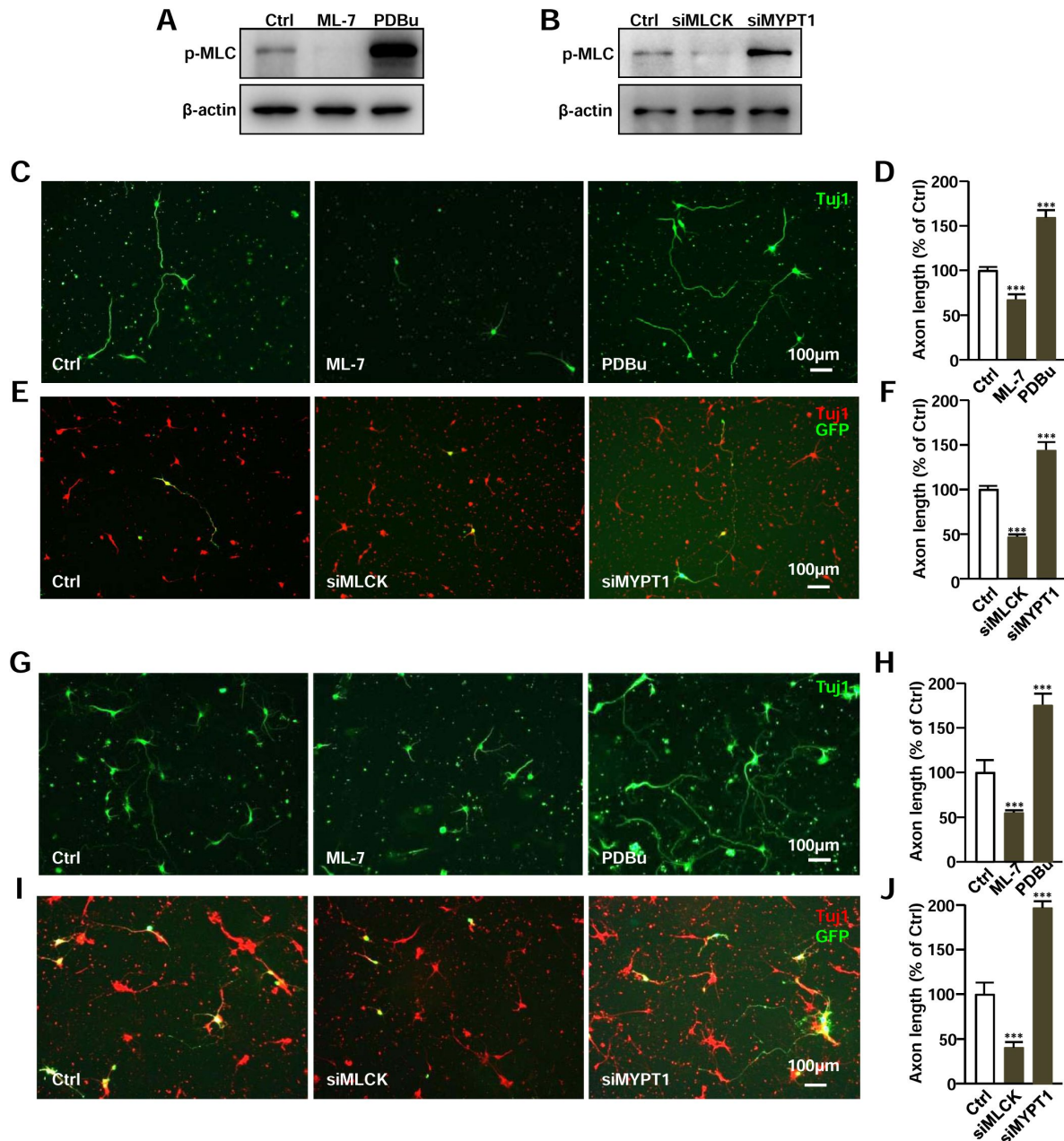


Figure 4.

Inhibition of MLCP activity promotes axon growth from developing embryonic neurons.

(A, B) Cortical neurons were isolated from embryonic day 14.5 embryos. Neurons were treated with DMSO, ML-7 or PDBu, or transfected with ctrl siRNA, siMLCK or siMYPT1. The p-MLC expression levels were examined using Western blot analysis after 3 days of culture.

(C-F) cortical neurons were treated with DMSO, ML-7 or PDBu or transfected with ctrl siRNA, siMLCK or siMYPT1. After three days, cells were stained with Tuj1 to visualize axons. Representative images (C, E) and quantification (D, F) of axonal length demonstrate ML-7 or siMLCK inhibits embryonic cortical neuronal axon growth. In contrast, inhibition of MLCP activity with PDBu or siMYPT1 promoted embryonic cortical neuronal axon growth ($N=3$, $***p < 0.001$). Scale bar: 100 μ m.

(G-J) Hippocampal neurons were isolated from embryonic day 18.5 embryos. Neurons were treated with DMSO, ML-7 or PDBu or transfected with scramble siRNA, siMLCK or siMYPT1. Similarly, Representative images (G, I) and quantification (H, J) of axonal length demonstrate that ML-7 or siMLCK inhibits hippocampal neuronal axon growth. In contrast, suppression of MLCP with PDBu or siMYPT1 promoted hippocampal neuronal axon growth ($N=3$, $***p < 0.001$). Scale bar: 100 μ m.

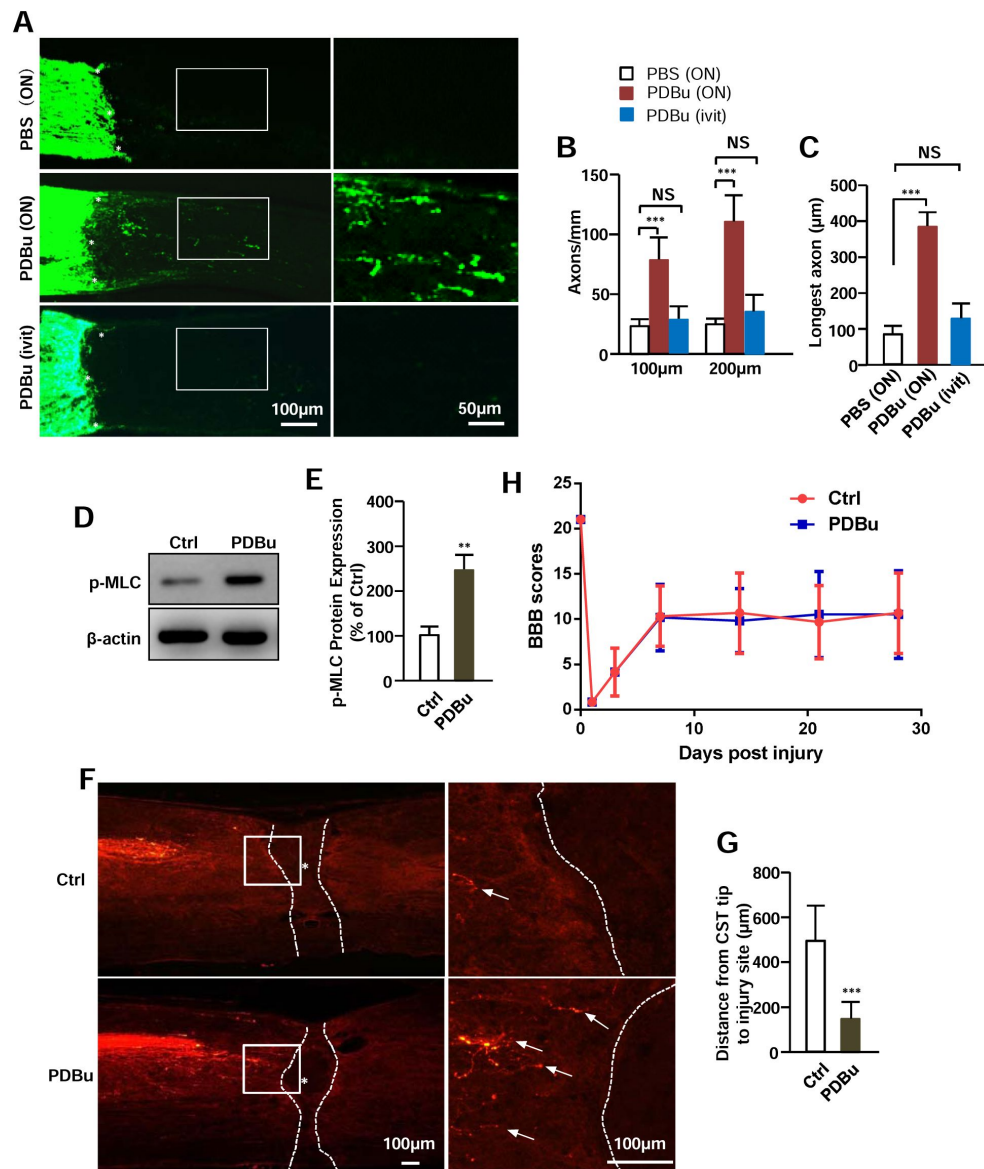


Figure 5.

Inhibition of MLCP activity promotes CNS axon regeneration in vivo.

(A-C) Local application of a gelatin sponge soaked with PBS (Ctrl) or 50.0 μ M PDBu on the site of the lesion (ON), or microinjection of PDBu (2.0 μ l, 50.0 μ M) into the vitreous body (ivit) were performed after optic nerve crush injury. Three day later, CTB-labeled regenerating axons were quantified. Representative images of optic nerve sagittal sections (A) and quantification of CTB-positive fibers (B) at the point distance 100 μ m and 200 μ m from the injury site indicate that only local administration of PDBu at the site of the lesion promotes the axon growth. (C) The longest axons per section of group were measured and quantified.

(D-H) T8 whole spinal cord crush injury was performed in adult mice then treated with PBS or 50.0 μ M PDBu by intrathecal injection once every 4 days. BDA was injected into the right sensorimotor cortex 2 weeks after spinal cord injury to label the regenerating cortical spinal tract (CST) axons. After 2 weeks, sagittal sections of the spinal cord around the crush site were created, and the distance between the tip of the axons and injury site was measured. Representative image of Western blots (D) and its quantification (E) demonstrates that intrathecal injection of PDBu increases p-MLC level, it means that PDBu sufficiently inhibits MLCP activity. Representative image of spinal cord sagittal section (F) and the measurement of distance from BDA-positive axon tips to the injury site demonstrate that intrathecal injections of PDBu prevent injury-induced axon retraction. Arrow indicates BDA-positive axons, dash line indicates lesion site. (H) Mean BBB scores evaluation indicated that there was no difference between PDBu treatment and the PBS control group. (N=5, ***p < 0.001).

The local inhibition of MLCP activity prevented axonal retraction in the injured spinal cord

After spinal cord injury, corticospinal tract (CST) axons also did not regenerate spontaneously, principally because of the inhibitory environment (Supplementary Figure S4). To test whether local inhibition of MLCP activity can also promote spinal cord axon regeneration, mice were subjected to spinal cord crush injury at the level of the eighth thoracic vertebra (T8)¹⁵. Briefly, the T8 spinal cord was exposed and crushed for 1 second with jeweler's forceps #5. A 100 μ L PDBu (50 μ M) or its vehicle, DMSO, was then administrated once every 4 days by intrathecal injection to block MLCP activity. After 2 weeks, 1.6 μ L of 10 % biotinylated dextran amine (BDA) was microinjected into the sensorimotor cortex to label regenerating axons¹⁶. After a further 2 weeks, BDA-labeled regenerating axon was evaluated in the spinal cord section. Intrathecal PDBu injection significantly boosted p-MLC levels (Figure 5D, E) and prevented injury-induced axon retraction (Figure 5F, G), while also greatly reducing the number of retraction bulbs (Supplementary Figure S5). However, we did not observe any BDA-labeled axon fibers beyond the lesion site, and Basso–Beattie–Bresnahan (BBB) score evaluation following PDBu treatment did not differ from that of the control group (Figure 5H). Together, these results indicated that the local inhibition of MLCP activity reduces injury-induced retraction bulb formation and axon retraction in the spinal cord.

MLCK may regulate axon regeneration independently of myosin II activity

We further examined whether the regulatory effect of MLCK/MLCP on axonal growth was mediated through myosin II activity. For this, dissociated DRG neurons were co-treated with blebbistatin (a pharmacological myosin II inhibitor) and ML-7. Consistent with our previous study¹, the inhibition of myosin II activity with 25 μ M blebbistatin markedly promoted axonal growth (Figure 6A, B). Surprisingly, however, ML-7/blebbistatin co-treatment did not block the inhibitory effect of ML-7 on sensory axon regeneration (Figure 6A, B). Additionally, blebbistatin administration did not influence MLC phosphorylation levels in cultured DRG neurons (Figure 6C, D). Collectively, these results interestingly suggested that MLCK regulates axon regeneration may independently of myosin II activity.

Local F-actin distribution in the growth cone is regulated by MLCK and MLCP

Axonal regeneration is requiring both regulation of gene expression in the soma and the cytoskeletal assembly at the growth cone. We previously established a culture-and-replating model to investigate whether axonal regeneration was soma gene transcription-dependent or local growth cone cytoskeleton assembly-dependent^{12,13}. Interestingly, only ML-7 treatment after replating was found to significantly inhibit axonal growth in peripheral sensory neurons. However, no effect was observed on axonal growth when ML-7 was administrated before cell replating (Figure 7A, B, C). Several studies have reported that peripheral axotomy triggers local cytoskeleton dynamics-dependent axon growth in DRG neurons^{13,16,20}. Consistent with the culture-and-replating results, axotomy-induced, local cytoskeleton assembly-dependent axon growth was also markedly inhibited by ML-7 treatment (Figure 7D, E). These results suggested that MLCK primarily influences local cytoskeleton dynamics during axonal growth. Accordingly, we further examined how MLCK/MLCP activity regulates growth cone cytoskeletal distribution during axonal growth. E14.5 cortical neurons were treated with ML-7 or PDBu for 3 days, following which microtubule and F-actin distribution in the growth cone was visualized using

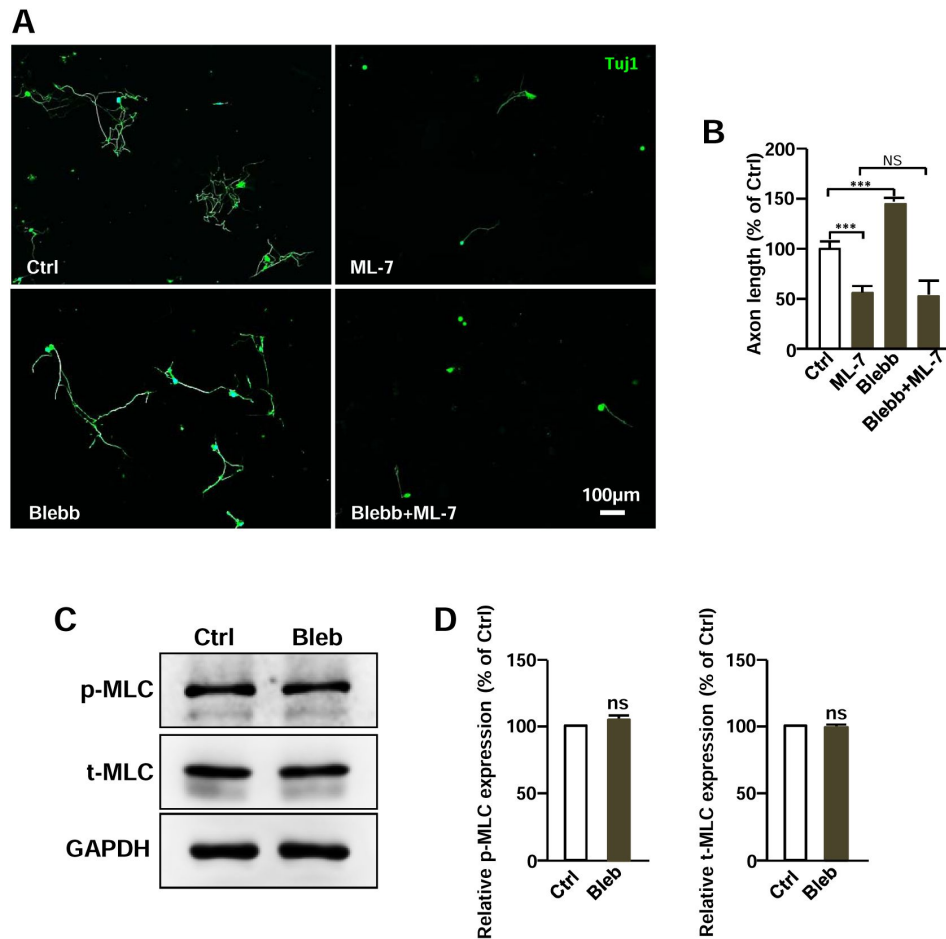


Figure 6.

MLCK regulates axon regeneration independent of myosin II activity.

(A, B) Adult L4-L5 DRG neurons were cultured *in vitro* for 3 days with the addition of the pharmacological inhibitors indicated. Representative images (A) and quantification of axon length (B) demonstrate that, although myosin II inhibitor blebbistatin markedly promoted sensory neuronal axon growth, however, it did not rescue the inhibitory effect of ML-7 on sensory axon growth (N=3, *** $p < 0.001$). (C, D) The blebbistatin administration did not influence MLC phosphorylation levels in cultured DRG neurons, representative image of Western blots (C) and its quantification (D) (N=3, ns: no significant).

Tuj1 and phalloidin staining. The results indicated that ML-7 treatment increased growth cone size and F-actin content, whereas PDBu treatment exerted the opposite effect (**Figure 8A-C**). These findings may indicate that MLCK and MLCP activity regulates F-actin redistribution.

Discussion

In the present study, we demonstrated that peripheral axotomy increases MLCK and MLC phosphorylation levels in adult DRG neurons subsequently. Thus, when MLCK activity was suppressed in mature DRG neurons *via* ML-7- or specific siRNA, axonal growth was blocked in treated neurons both *in vitro* and *in vivo*. Conversely, an inhibit in MLCP activity *via* the knockdown of MYPT1 significantly promoted axonal growth in both PNS and CNS neurons. We further found that the MLCK and MLCP primarily regulates F-actin redistribution in the growth cone during the axon regeneration process.

Axon injury leads to disconnection between neurons and their targets, thereby destroying functional circuits and behavioral outputs. This implies that for the successful restoration of impaired nervous system function, injured axons must regenerate to their original targets and reconstruct functional circuits. Interestingly, it has been reported that the coordinated regulation of the neuronal cytoskeleton dynamics is essential for axon shaft extension²¹. Among them, the microtubule and actin are two of major component of neuronal cytoskeleton that involves axonal extension. For example, Athamneh et al. found that the speed of axon elongation is highly correlated with the bulk forward translocation rate of microtubules²². In addition, it also has been reported that the axon extension depends on actin assembly and microtubule-actin interactions²³. Unsurprisingly given these observations, increasing attention has been paid to promoting axon regeneration *via* the modulation of local cytoskeletal dynamics in the growth cone. Myosin II is a key cytoskeleton protein, which activated through the MLCK-mediated phosphorylation of its light chain and deactivated *via* the dephosphorylation of the N-terminus of its light chain by MLCP. Here, we found that the MLCK expression and phosphorylation level of MLC was increased in adult DRG neurons after sciatic nerve injury, while the inhibition of MLCK blocked axon regeneration. These results indicate that activated MLCK is required for mammalian axon regeneration. If thus, increase of MLCK expression will activate myosin II during axon regeneration. However, in our previous study, we found that pharmacological blockade of myosin II activity with blebbistatin stimulated axon growth¹, which suggested that myosin II plays an inhibitory role in this process. This apparent contradiction might be attributed to blebbistatin being a specific inhibitor of non-muscle myosin II ATPase activity and may not affect MLCK activity or the phosphorylation status of MLC²⁴. Indeed, blebbistatin treatment did not influence MLC phosphorylation levels in our study, in line with a previous report in which it was found that blebbistatin does not alter the phosphorylation status of MLC in sensory neurons and trabecular meshwork cells^{24,35}. Additionally, our results demonstrated that blebbistatin treatment could not rescue the inhibitory effect of ML-7 on axon regeneration. These data suggest that the regulatory effect of MLCK on axonal growth may be exerted through different molecular mechanisms. Interestingly, it has been reported that MLCK regulates cell migration not by MLC phosphorylation, but possible MLCK may serve as an F-actin-binding protein stabilizing the F-actin/myosin II network of the membrane cytoskeleton²⁵. Additionally, recent study also identified multiple noncanonical targets of MLCK through CRISPR-Cas9/phosphoproteomics method²⁶, thus it is also possible that MLCK regulates axon growth its noncanonical targets or unknown mechanism. They also found that MLCK plays an important role in vasopressin-induced actin depolymerization²⁶. Studies have shown that the combination of MLCK with F-actin is critical to its function. Short-chain MLCK binds to F-actin via three DFRXXL motifs at its N-terminal, while long-chain MLCK has two additional DFRXXL motifs and six IG-like modules, which may give it unique binding properties in cell localization²⁵. During axon growth, the dynamic changes of F-actin are regulated by a variety of proteins, such as ADF/cofilin, Arp2/3, Eps8, Profilin, myosin II, myosin V, etc. These proteins regulate axon growth and orientation by

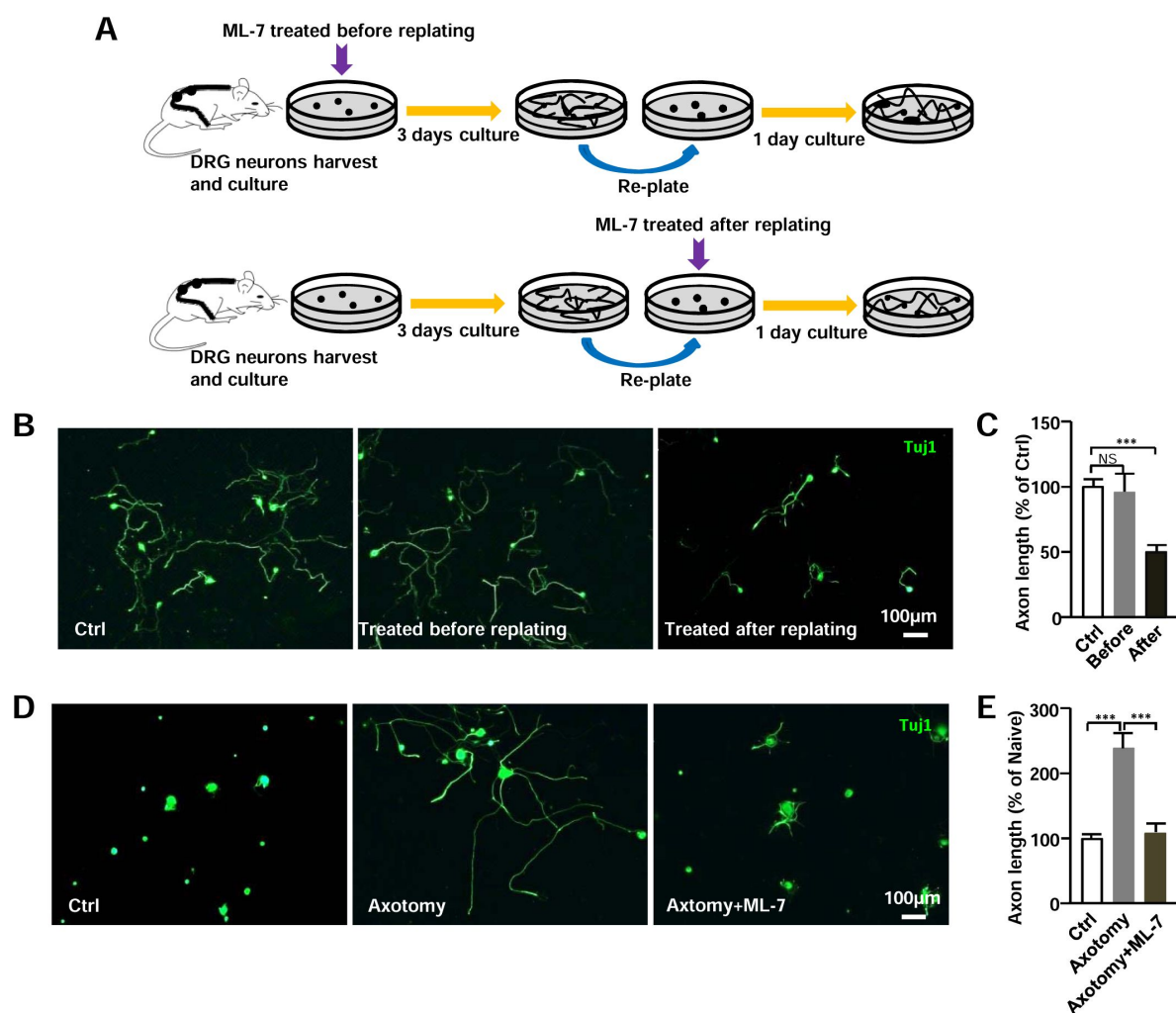


Figure 7.

Inhibition of MLCK blocks transcription-independent axon growth of adult sensory neurons.

(A) Schematic figure of the culture-and-replate model. Adult sensory neurons were cultured for 3 days then replated to allow growth of new axons for one day. During this process, ML-7 was added prior to or after replating.

(B, C) Representative images (B) and quantification of axon length (C) indicate that treatment with ML-7 after replating significantly inhibits adult sensory axon growth. However, treatment with ML-7 prior to replating has no effect on adult sensory neural axon growth ($N=3$, $***p < 0.001$).

(D, E) Mice underwent sciatic nerve axotomy 7 days previously. L4-L5 DRGs were harvested from naive or axotomized mice and cultured for 1 day with ML-7 treatment. Neurons were stained with Tuj1. Representative images (D) and quantification of axon length (E) indicate that blockade of MLCK activity with ML-7 inhibits axotomy-induced axon growth of adult sensory neurons ($N=3$, $***p < 0.001$).

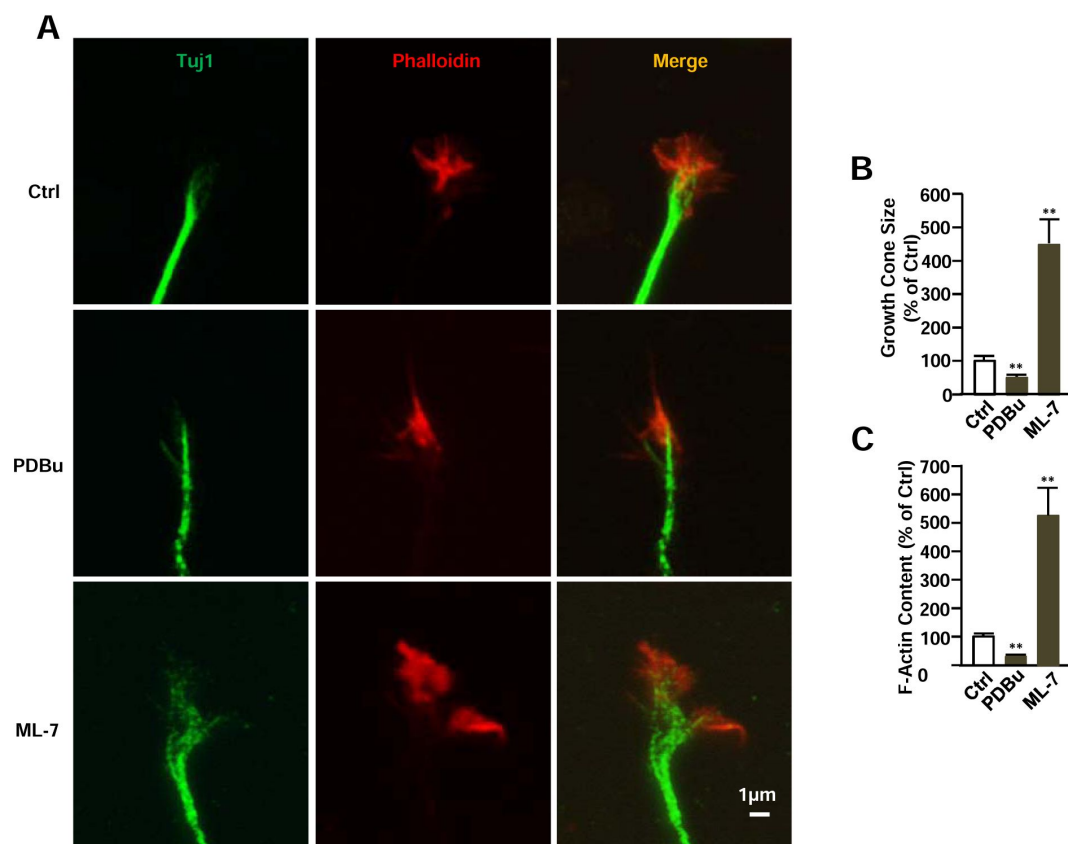


Figure 8.

MLCK and MLCP activity regulate F-actin distribution of the growth cone.

(A, B) Embryonic E14.5 cortical neurons were isolated and treated with DMSO, PDBu or ML-7. After 3 days of culture, neurons were stained with Tuj1 antibody (Green) to visualize tubulin and phalloidin (Red) to visualize F-actin. Representative images (A), quantification of the size of the growth cone (B), and F-actin content (C) indicate that ML-7 treatment increases the growth cone size and F-actin content, and PDBu treatment reduces the growth cone size and the F-actin content. Scale bar: 1µm (N=3, *** $p < 0.001$).

influencing the polymerization and depolymerization of F-actin. For example, cofilin increases the availability of actin by promoting the depolymerization of F-actin, which promotes axon growth³⁶. Furthermore MLCP inhibition also increase actin arc movement in growth cone⁵. This means that MLCK/MLCP may also regulates axon growth via actin dynamics. In addition, other study also has shown that the absence of MLCK not only impairs the synthesis of transmembrane complexes, such as integrin, and fibronectin, but also disrupts the balance of membrane tension and protrusion, thereby causing the membrane cytoskeleton instability^{37,38}. We think these may affect axon growth. Meanwhile, adult sensory neurons express different isoforms of myosin II, which also exert distinct effects on axonal regeneration²⁷. For example, the knockdown of myosin IIA enhances axonal growth, while the knockdown of myosin IIB increases F-actin retrograde flow and inhibits axon growth^{1,28–31}. A study in Neuro-2A cells also revealed that myosin IIA is involved in axon retraction, whereas myosin IIB was reported to contribute to axon extension^{28,29}. Interesting in neurons, although expression of both NMIIA and NMIIIB has been reported, the NMIIIB being the predominant one³⁹. Under physiological conditions, MLCP is activated by PKC, and it has been reported that the PKC-mediated activation of myosin II is restricted to the T zone of growth cones; the results of the present study suggest that growth cones require low-to-moderate levels of myosin II activity and F-actin for best growth rates. This observation emphasizes the complexity of the role of myosin II in axonal growth, the precise molecular mechanisms of which require further in-depth investigation.

The inability of neurons in the CNS to regenerate their axons after injury is attributable, at least in part, to the presence of inhibitory molecules at the injury site, including CSPGs and myelin. Here, we found that the inhibition of MLCP activity with PDBu or the siRNA-mediated knockdown of MYPT1 promoted axonal growth in adult sensory neurons over both CSPG and myelin substrates. Our findings also further indicated that the local inhibition of MLCP by PDBu not only induces axon regeneration in peripheral sensory neurons but also promotes axon regeneration in the optic nerve. Studies have shown that MLCK possesses several actin-binding domains to which F-actin can bind and form bundles^{32,33} and that directional growth cone motility during axon growth is controlled by actin-based machinery. Growth cone movement during axon growth is achieved through protrusion towards attractive guidance cues and retraction from repulsive ones in an actin-dependent manner. In the classic molecular clutch model, strong mechanical coupling between the F-actin cytoskeleton and the substrate, mediated by cell adhesion complexes, transmits the forces generated by the cytoskeleton into rearward traction forces, enabling the forward movement of the growth cone. Our results indicated that treating DRG neurons with ML-7 induces the increases F-actin content in the growth cone. Conversely, the inhibition of MLCP following PDBu administration resulted in decreased F-actin content. Our findings are consistent with the results of a previous study on *Helisoma*, in which treatment with ML-7 increased F-actin content at the C/P-domain interface, thus inhibiting axonal regeneration³⁴. This suggests that MLCK or MLCP may control mammalian axonal regeneration by regulating growth cone F-actin distribution.

Together, our findings indicate that directly targeting growth cone cytoskeleton components may be a promising strategy for promoting mammalian axon regeneration.

Statements & declarations

Funding

This work was supported by the National Natural Science Foundation of China (Nos. 81571189 to S), A Priority Academic Program Development of Jiangsu Higher Education Institutions, and Innovation and Entrepreneurship Program of Jiangsu Province.

Author Contribution

Saijilafu, W.H. W and J.J. M designed the experiment. W.H. W and J.J. M performed the experiments and analyzed the data. W.H. W, Y.Y. Y.X. M and Saijilafu co-wrote the paper with all authors' input.

Ethics approval

The animal experiments were conducted in accordance with the National Institutes of Health Guide for the Care and Use of Laboratory Animals (NIH Publications No. 8023, revised 1978) and were approved by the Ethics Committee of the First Affiliated Hospital of Soochow University.

Consent to publish

All the authors consent for publication.

Conflict of interests

The authors have no relevant financial interests to disclose.

Data availability

All data generated or analyzed during this study are available from the corresponding author upon reasonable request.

Figure S1.

ML-7 treatment inhibits axon growth in a dose-dependent manner.

The DRG neurons from adult mice were cultured and treated with a series of doses of ML-7. After 3 days culture, neurons were stained with Tuj1 antibody and axonal lengths quantified.

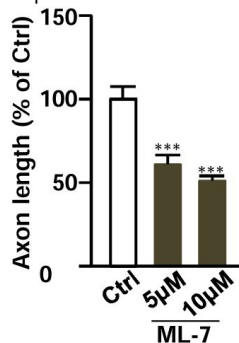
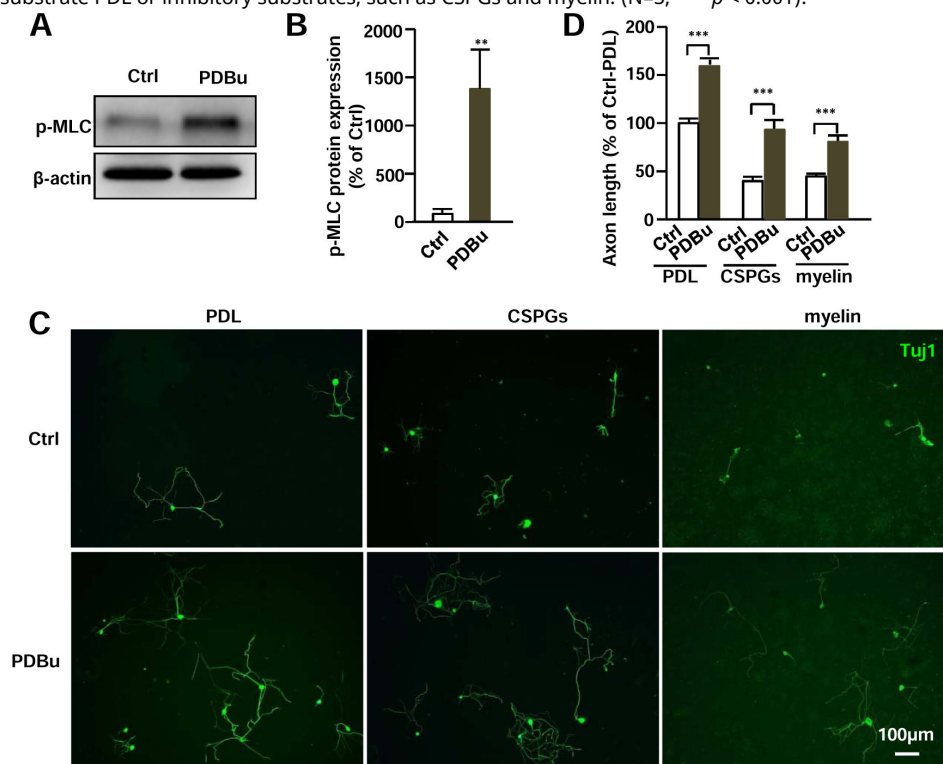


Figure S2.

PDBu treatment promotes axon growth from adult peripheral sensory neurons.

(A, B) Representative Western blot image (A) and quantification of protein band (B) indicates that administration of PDBu, the MLCP pharmacological inhibitor, in cultured DRG neurons increases p-MLC expression (N=3, ** $p < 0.01$).

(C, D) DRG neurons were cultured on permissive (PDL) or inhibitory substrates (CSPGs or myelin) in the presence or absence of PDBu, as indicated. Neurons were stained with Tuj1. Representative images (C) and quantification of axon length (D) show that increased p-MLC expression with PDBu addition, which promotes adult sensory neuronal axon growth on either permissive substrate PDL or inhibitory substrates, such as CSPGs and myelin. (N=3, *** $p < 0.001$).



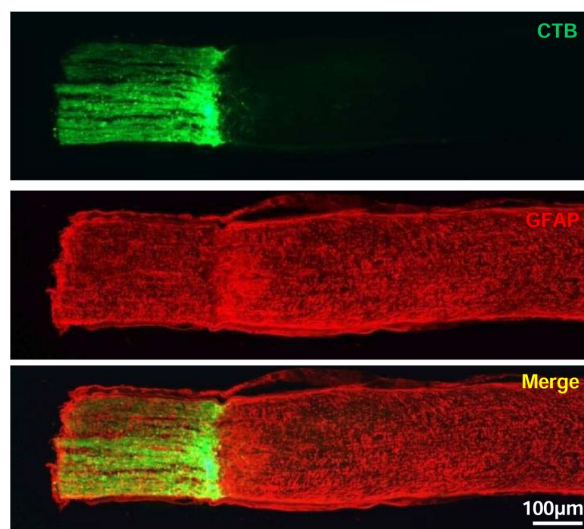


Figure S3.

Optic nerve crush injury model

Optic nerve was crushed behind the eyeball with a custom-modified jeweler's 5[#] forceps for 1 second. Two weeks later, 2.0 µl Alexa Fluor 488-conjugated CTB (green) was injected intravitreally to label regenerating optic nerve axons.

Immunofluorescent staining of GFAP staining (red) was also performed in the longitudinal section of crushed optic nerves.

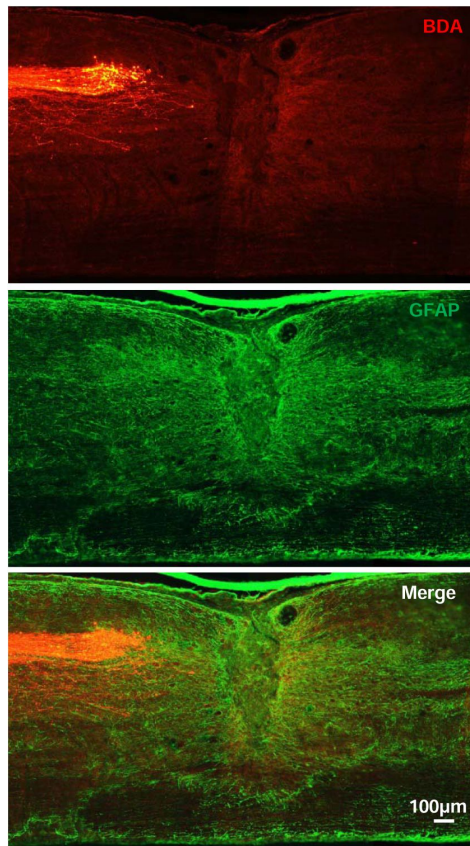


Figure S4.

Whole spinal cord crush injury model

Whole spinal cord was crushed at T8 level with a custom-modified jeweler's 5[#] forceps for 1 second. Two weeks later, a 1.6 μ l aliquot of 10% BDA was injected into the sensorimotor cortex to label regenerating cortical spinal tract axons, and another two weeks later, the spinal cord was dissected out. Immunofluorescent staining of BDA (red) and GFAP staining (green) was performed in the sagittal section of spinal cord.

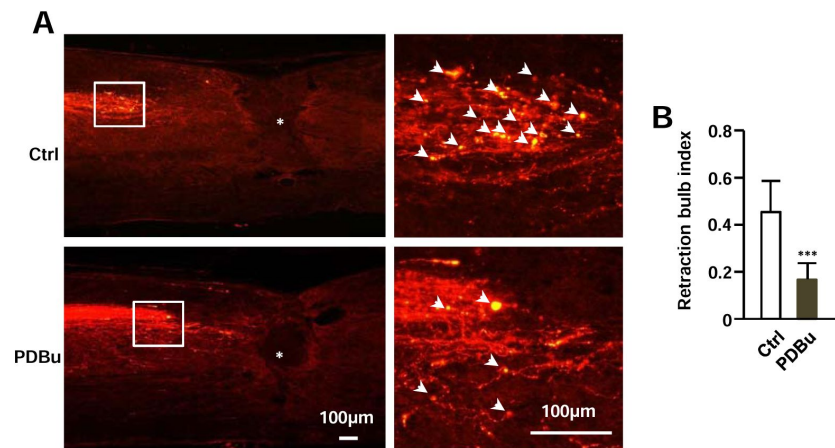


Figure S5.

Intrathecal PDBu injection prevented axonal retraction bulb formation.

Representative images of spinal cord sagittal section (A) and quantification of retraction bulb formation (B). White triangle indicates retraction bulb (N=3, *** $p < 0.001$).

References

1. Hur E.-M., et al. (2011) **Engineering neuronal growth cones to promote axon regeneration over inhibitory molecules** *Proceedings of the National Academy of Sciences* **108**:5057–5062
2. Sengottuvel V., Leibinger M., Pfreimer M., Andreadaki A., Fischer D (2011) **Taxol Facilitates Axon Regeneration in the Mature CNS** *The Journal of Neuroscience* :2688–2699 <https://doi.org/10.1523/jneurosci.4885-10.2011>
3. Hellal F., et al. (2011) **Microtubule Stabilization Reduces Scarring and Causes Axon Regeneration After Spinal Cord Injury** *Science* **331**:928–931
4. Riviaccio M. A., et al. (2009) **HDAC6 is a target for protection and regeneration following injury in the nervous system** *Proceedings of the National Academy of Sciences* **106**:19599–19604
5. Zhang X.-F., Schaefer A. W., Burnette D. T., Schoonderwoert V. T., Forscher P (2003) **Rho-Dependent Contractile Responses in the Neuronal Growth Cone Are Independent of Classical Peripheral Retrograde Actin Flow** *Neuron* **40**:931–944
6. Schaefer A. W., Kabir N., Forscher P (2002) **Filopodia and actin arcs guide the assembly and transport of two populations of microtubules with unique dynamic parameters in neuronal growth cones** *The Journal of Cell Biology* **158**:139–152
7. Jian X., Szaro B. G., Schmidt J. T (1996) **Myosin light chain kinase: expression in neurons and upregulation during axon regeneration** *Journal of Neurobiology* **31**:379–391
8. Kim Y.-S., Fritz J. L., Seneviratne A. K., VanBerkum M. F. A (2002) **Constitutively Active Myosin Light Chain Kinase Alters Axon Guidance Decisions in Drosophila Embryos** *Developmental Biology* **249**:367–381
9. Zhang X.-F., Hyland C., Van Goor D., Forscher P (2012) **Calcineurin-dependent cofilin activation and increased retrograde actin flow drive 5-HT-dependent neurite outgrowth in Aplysia bag cell neurons** *Molecular Biology of the Cell* **23**:4833–4848
10. Tao T., et al. (2017) **Minocycline Promotes Neurite Outgrowth of PC12 Cells Exposed to Oxygen-Glucose Deprivation and Reoxygenation Through Regulation of MLCP/MLC Signaling Pathways** *Cellular and Molecular Neurobiology* **37**:417–426
11. Minor K., et al. (2008) **Decorin promotes robust axon growth on inhibitory CSPGs and myelin via a direct effect on neurons** *Neurobiology of Disease* **32**:88–95
12. Saijilafu & Zhou, F. Q (2012) **Genetic Study of Axon Regeneration with Cultured Adult Dorsal Root Ganglion Neurons** *Journal of Visualized Experiments* <https://doi.org/10.3791/4141>
13. Saijilafu, et al. (2013) **PI3K-GSK3 signalling regulates mammalian axon regeneration by inducing the expression of Smad1** *Nature Communications* **4**
14. Saijilafu Hur E. M., Zhou F. Q. (2011) **Genetic dissection of axon regeneration via in vivo electroporation of adult mouse sensory neurons** *Nature Communications* <https://doi.org/10.1038/ncomms1568>

15. Liu K., et al. (2010) **PTEN deletion enhances the regenerative ability of adult corticospinal neurons** *Nature Neuroscience* :1075–1081 <https://doi.org/10.1038/nn.2603>
16. Xi F., et al. (2019) **Calcium/calmodulin-dependent protein kinase II regulates mammalian axon growth by affecting F-actin length in growth cone** *Journal of Cellular Physiology* **234**:23053–23065
17. Basso D M., Beattie M. S., Bresnahan J. C. (1995) **A Sensitive and Reliable Locomotor Rating Scale for Open Field Testing in Rats** *Journal of Neurotrauma* **12**:1–21
18. El-Yazbi A. F., Abd-Elrahman K. S., Moreno-Dominguez A (2015) **PKC-mediated cerebral vasoconstriction: Role of myosin light chain phosphorylation versus actin cytoskeleton reorganization** *Biochemical Pharmacology* **95**:263–278
19. Shirazi A., et al. (1994) **Purification and characterization of the mammalian myosin light chain phosphatase holoenzyme. The differential effects of the holoenzyme and its subunits on smooth muscle** *Journal of Biological Chemistry* **269**:31598–31606
20. Smith D. S., Pate Skene J. H (1997) **A Transcription-Dependent Switch Controls Competence of Adult Neurons for Distinct Modes of Axon Growth** *The Journal of Neuroscience* **17**:646–658
21. Poulain F. E., Sobel A (2010) **The microtubule network and neuronal morphogenesis: Dynamic and coordinated orchestration through multiple players** *Molecular and Cellular Neuroscience* **43**:15–32
22. (2017) Athamneh, A. I. M., et al. **Neurite elongation is highly correlated with bulk forward translocation of microtubules.** *Scientific Reports* **7**, (2017).
23. Miller K. E., Suter D. M (2018) **An Integrated Cytoskeletal Model of Neurite Outgrowth** *Frontiers in Cellular Neuroscience* **12**
24. Yu P., Santiago L. Y., Katagiri Y., Geller H. M (2012) **Myosin II activity regulates neurite outgrowth and guidance in response to chondroitin sulfate proteoglycans** *Journal of Neurochemistry* **120**:1117–1128
25. Chen C., et al. (2014) **Myosin Light Chain Kinase (MLCK) Regulates Cell Migration in a Myosin Regulatory Light Chain Phosphorylation-independent Mechanism** *Journal of Biological Chemistry* **289**:28478–28488
26. Isobe K., et al. (2020) **CRISPR-Cas9/phosphoproteomics identifies multiple noncanonical targets of myosin light chain kinase** *American Journal of Physiology-Renal Physiology* :F600–F616 <https://doi.org/10.1152/ajprenal.00431.2019>
27. Bridgman P.C, Elkin L.L (2000) **Axonal myosins** *Journal of Neurocytology* **29**:831–41
28. Wylie S. R., Chantler P. D (2003) **Myosin IIA Drives Neurite Retraction** *Molecular Biology of the Cell* **14**:4654–4666
29. Wylie S. R., Wu P., Patel H., Chantler P. D (1998) **A conventional myosin motor drives neurite outgrowth** *Proceedings of the National Academy of Sciences* **95**:12967–12972
30. Turney S. G., Bridgman P. C (2005) **Laminin stimulates and guides axonal outgrowth via growth cone myosin II activity** *Nature Neuroscience* **8**:717–719

31. Bridgman P. C., Dave S., Asnes C. F., Tullio A. N., Adelstein R. S (2001) **Myosin IIB Is Required for Growth Cone Motility** *The Journal of Neuroscience* **21**:6159–6169
32. Hayakawa K., Okagaki T., Higashifujime S., Kohama K (1994) **Bundling of Actin Filaments by Myosin Light Chain Kinase from Smooth Muscle** *Biochemical and Biophysical Research Communications* **199**:786–791
33. Sellers J. R., Pato M. D (1984) **The binding of smooth muscle myosin light chain kinase and phosphatases to actin and myosin** *Journal of Biological Chemistry* **259**:7740–7746
34. Torrealan P. J., Waterman-Storer C. M., Cohan C. S (2005) **The effects of collapsing factors on F-actin content and microtubule distribution of Helisoma growth cones** *Cell Motility and the Cytoskeleton* **60**:166–179
35. Zhang M., Rao PV (2005) **Blebbistatin, a novel inhibitor of myosin □ ATPase activity, increases aqueous humor outflow facility in perfused enucleated porcine eyes** *Investigative ophthalmology & visual science* **46**:4130–4138
36. Tedeschi Andrea *et al.* (2019) **ADF/Cofilin-Mediated Actin Turnover Promotes Axon Regeneration in the Adult CNS** *Neuron* **103**:1073–1085
37. Yang C.X., Chen H.Q., Chen C., Yu W.P., Zhang W.C., Peng Y.J., He W.Q., Wei D.M., Gao X., Zhu M.S. (2006) **Microfilament-binding properties of n-terminal extension of the isoform of smooth muscle long myosin light chain kinase** *Cell Res* **16**:367–376
38. Gao Y., Ye L.H., Kishi H., Okagaki T., Samizo K., Nakamura A., Kohama K. (2001) **Myosin light chain kinase as a multifunctional regulatory protein of smooth muscle contraction** *Iubmb Life* **51**:337–344
39. Costa Ana Rita, Sousa Monica M (2020) **Non-Muscle Myosin II in Axonal Cell Biology: From the Growth Cone to the Axon Initial Segment** *Cells* **9**

Author information

Saijilafu

Key Laboratory of Novel Targets and Drug Study for Neural Repair of Zhejiang Province, School of Medicine, Hangzhou City University, Hangzhou, China, Department of Orthopaedics, the First Affiliated Hospital, Orthopaedic Institute, Soochow University, Suzhou, China
ORCID iD: [0000-0003-2290-9808](https://orcid.org/0000-0003-2290-9808)

For correspondence: saijlf@hzcw.edu.cn

Wei-Hua Wang

Department of Orthopaedics, the First Affiliated Hospital, Orthopaedic Institute, Soochow University, Suzhou, China

Jin-Jin Ma

Department of Orthopaedics, the First Affiliated Hospital, Orthopaedic Institute, Soochow University, Suzhou, China

Yin Yin

Laboratory Animal Center, Suzhou Medical College of Soochow University, Suzhou, China

Yan-Xia Ma

Department of Orthopaedics, the First Affiliated Hospital, Orthopaedic Institute, Soochow University, Suzhou, China

Editors

Reviewing Editor

Nicolas Unsain

INIMEC-CONICET - Universidad Nacional de Córdoba, Córdoba, Argentina

Senior Editor

K VijayRaghavan

National Centre for Biological Sciences, Tata Institute of Fundamental Research, Bangalore, India

Reviewer #1 (Public review):

This paper examines the role of MLCK (myosin light chain kinase) and MLCP (myosin light chain phosphatase) in axon regeneration. Using loss-of-function approaches based on small molecule inhibitors and siRNA knockdown, the authors explore axon regeneration in cell culture and in animal models from central and peripheral nervous systems. Their evidence shows that MLCK activity facilitates axon extension/regeneration, while MLCP prevents it.

Major concerns:

(1) In the title, authors indicate that the observed effects from loss-of-function of MLCK/MLCP take place via F-actin redistribution in the growth cone. However, there are no experiments showing a causal effect between changes in axon growth mediated by MLCK/MLCP and F-actin redistribution.

(2) The author combines MLCK inhibitors with Bleb (Figure 6), trying to verify if both pairs of inhibitors act on the same target/pathway. MLCK may regulate axon growth independent of NMII activity. However, this has very important implications for the understanding not only on how NMII works and affects axon extension, but also in trying to understand what MLCP is doing. One wonders if MLCP actions, which are opposite of MLCK, also independent of NMII activity? The authors, in the discussion section, try to find an explanation for this finding, but I consider it fails since the whole rationale of the manuscript is still around how MLCK and MLCP affect NMII phosphorylation.

What follows is a discussion of the merits and limitations of different claims of the manuscript in light of the evidence presented.

(1) Using western blot and immunohistochemical analyses, authors first show that MLCK expression is increased in DRG sensory neurons following peripheral axotomy, concomitant to an increase in MLC phosphorylation, suggesting a causal effect (Figure 1). The authors claim that it is common that axon growth-promoting genes are upregulated. It would have been interesting at this point to study in this scenario the regulation of MLCP.

(2) Using DRG cultures and sciatic nerve crush in the context of MLCK inhibition (ML-7) and down-regulation, authors conclude that MLCK activity is required for mammalian peripheral axon regeneration both in vitro and in vivo (Figure 2). In parallel, the authors show that these treatments affect as expected the phosphorylation levels of MLC.

The in vitro evidence is of standard methods and convincing. However, here, as well as in all other experiments using siRNAs, no Control siRNAs were used. Authors do show that the target protein is downregulated, and they can follow transfected cells with GFP. Still, it should be noted that the standard control for these experiments has not been done.

(3) The authors then examined the role of the phosphatase MLCP in axon growth during regeneration. The authors first use a known MLCP blocker, phorbol 12,13-dibutyrate (PDBu), to show that is able to increase the levels of p-MLC, with a concomitant increase in the extent of axon regrowth of DRG neurons, both in permissive as well as non-permissive substrates. The authors repeat the experiments using the knockdown of MYPT1, a key component of the MLC-phosphatase, and again can observe a growth-promoting effect (Figure 3).

The authors further show evidence for the growth-enhancing effect in vivo, in nerve crush experiments. The evidence in vivo deserves more evidence and experimental details (see comment 2). A key weakness of the data was mentioned previously: no control siARN was used.

(4) In the next set of experiments (presented in Figure 4) authors extend the previous observations in primary cultures from the CNS. For that, they use cortical and hippocampal cultures, and pharmacological and genetic loss-of-function using the above-mentioned strategies. The expected results were obtained in both CNS neurons: inhibition or knockdown of the kinase decreases axon growth, whereas inhibition or knockdown of the phosphatase increases growth. A main weakness in this set is that drugs were used from the beginning of the experiment, and hence, they would also affect axon specification. As pointed in Materials and Method (lines 143-145) authors counted as "axons" neurites longer than twice the diameter of the cell soma, and hence would not affect the variable measured. In any case, to be sure one is only affecting axon extension in these cells, the drugs should have been used after axon specification and maturation, which occurs at least after 5 DIV.

(5) In Figure 7, the authors a local cytoskeletal action of the drug, but the evidence provided does not differentiate between a localized action of the drugs and a localized cell activity.

References:

- (1) Eun-Mi Hur 1, In Hong Yang, Deok-Ho Kim, Justin Byun, Saijilafu, Wen-Lin Xu, Philip R Nicovich, Raymond Cheong, Andre Levchenko, Nitish Thakor, Feng-Quan Zhou. 2011. Engineering neuronal growth cones to promote axon regeneration over inhibitory molecules. *Proc Natl Acad Sci U S A*. 2011 Mar 22;108(12):5057-62. doi: 10.1073/pnas.1011258108.
- (2) Garrido-Casado M, Asensio-Juárez G, Talayero VC, Vicente-Manzanares M. 2024. Engines of change: Nonmuscle myosin II in mechanobiology. *Curr Opin Cell Biol*. 2024 Apr;87:102344. doi: 10.1016/j.ceb.2024.102344.
- (3) Karen A Newell-Litwa 1, Rick Horwitz 2, Marcelo L Lamers. 2015. Non-muscle myosin II in disease: mechanisms and therapeutic opportunities. *Dis Model Mech*. 2015 Dec;8(12):1495-515. doi: 10.1242/dmm.022103.

<https://doi.org/10.7554/eLife.101250.2.sa2>

Reviewer #2 (Public review):

Summary:

Saijilafu et al. demonstrate that MLCK/MLCP proteins promote axonal regeneration in both the central nervous system (CNS) and peripheral nervous system (PNS) using primary cultures of adult DRG neurons, hippocampal and cortical neurons, as well as in vivo

experiments involving sciatic nerve injury, spinal cord injury, and optic nerve crush. The authors show that axon regrowth is possible across different contexts through genetic and pharmacological manipulation of these proteins. Additionally, they propose that MLCK/MLCP may regulate F-actin reorganization in the growth cone, which is significant as it suggests a novel strategy for promoting axonal regeneration.

Strengths:

This manuscript presents a wide range of experimental models to address its hypothesis and biological question. Notably, the use of multiple in vivo models significantly enhances the overall validity of the study.

Weaknesses:

- The authors previously published that blocking myosin II activity stimulates axonal growth and that MLCK activates myosin II. The present work shows that inhibiting MLCK blocks axonal regeneration while blocking MLCP (the protein that dephosphorylates myosin II) produces the opposite effect. Although this contradiction is discussed, no new evidence has been added to the manuscript to clarify this mechanism or address the remaining questions. Critical unresolved questions include: what happens to myosin II expression when both MLCK and MLCP are inhibited? If MLCK/MLCP are acting through an independent mechanism, what would that mechanism be?
- In the discussion, the authors mention the existence of two myosin II isoforms with opposing effects on axonal growth. Still, there is no evidence in the manuscript to support this point.
- It is also unclear how MLCK/MLCP acts on the actin cytoskeleton. The authors suggest that proteins such as ADF/cofilin, Arp 2/3, Eps8, Profilin, Myosin II, and Myosin V could regulate changes in F-actin dynamics. However, this study provides no experimental evidence to determine which proteins may be involved in the mechanism.

<https://doi.org/10.7554/eLife.101250.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

This paper examines the role of MLCK (myosin light chain kinase) and MLCP (myosin light chain phosphatase) in axon regeneration. Using loss-of-function approaches based on small molecule inhibitors and siRNA knockdown, the authors explore axon regeneration in cell culture and in animal models. Their evidence shows that MLCK activity facilitates axon extension/regeneration, while MLCP prevents it.

Major concern:

A global inconsistency in the conclusions of the authors is evident when trying to understand the role of NMII in axon growth and to understand the present results in light of previous reports by the authors and many others on the role of NMII in axon extension. The discussion of the matter fails to acknowledge a vast literature on how NMII activity is regulated. The authors study enzymes responsible for the phosphorylation and dephosphorylation of NMII, referring to something that is strongly proven elsewhere, that phosphorylation activates NMII and dephosphorylation deactivates it. The authors mention their own previous evidence using inhibitors of NMII ATPase activity (blebbistatin, Bleb for short) and inhibitors of a kinase that

phosphorylates NMII (ROCK), highlighting that Bleb increases axon growth. Since Bleb inhibits the ATPase activity of NMII, it follows that NMII is in itself an inhibitor of axon growth, and hence when NMII is inhibited, the inhibition on axon growth is relieved, and axonal growth takes place (REF1). It is known that NMII exists in an inactive folded state, and ser19 phosphorylation (by MLCK or ROCK) extends the protein, allowing NMII filament formation, ATPase activity, and force generation on actin filaments (REF2). From this, it is derived that if MLCK is inhibited, then there is no NMII phosphorylation, and hence no NMII activity, and, according to their previous work, this should promote axon growth. On the contrary, the authors show the opposite effect: in the lack of phospho-MLC, authors show axon growth inhibition.

We thank the Reviewer for taking time to review our manuscript, and we really appreciated the comments from the reviewer. We have tried our best to revise the manuscript to address all the comments raised by the Reviewer.

Reporting evidence challenging previous conclusions is common business in scientific endeavors, but the problem with the current manuscript is that it fails to point to and appropriately discuss this contradiction. Instead, the authors refer to the fact that MLCK and Bleb inhibit NMII in different steps of the activation process. While this is true, this explanation does not solve the contradiction. There are many options to accommodate the information, but it is not the purpose of this revision to provide them. Since the manuscript is focused solely on phosphorylation states of MLC and axon extension, the claims are simply at odds with the current literature, and this important finding, if true, is not properly discussed.

Thank you for reviewer's very good comments. As suggested by Reviewer, we discuss more detail it in our revised manuscripts (line 357-368; line 373-374).

What follows is a discussion of the merits and limitations of different claims of the manuscript in light of the evidence presented.

(1) Using western blot and immunohistochemical analyses, authors first show that MLCK expression is increased in DRG sensory neurons following peripheral axotomy, concomitant to an increase in MLC phosphorylation, suggesting a causal effect (Figure 1). The authors claim that it is common that axon growth-promoting genes are upregulated. It would have been interesting at this point to study in this scenario the regulation of MLCP, which is a main subject in this work, and expect its downregulation.

We thank the Reviewer for taking time to review our manuscript, and we really appreciated the positive comments from the Reviewer.

(2) Using DRG cultures and sciatic nerve crush in the context of MLCK inhibition and down-regulation, authors conclude that MLCK activity is required for mammalian peripheral axon regeneration both in vitro and in vivo (Figure 2).

The in vitro evidence is of standard methods and convincing. However, here, as well as in all other experiments using siRNAs, it is not clear what the control is about (the identity of the plasmids and sequences, if any).

We used the pCMV-EGFP-N3 as control, and the pCMV-EGFP-N3 plasmid was from Clontech, Inc. (line 114-115).

Related to this, it is not helpful to show the same exact picture as a control example in Figures 2 and 3 (panels J and E, respectively). Either because they should not have received the same control treatment, or simply because it raises concern that there are

no other control examples worth showing. In these images, it is not also clear where and how the crush site is determined in the GFP channel. This is of major importance since the axonal length is measured from the presumed crush site. Apart from providing further details in the text, the authors should include convincing images.

Thank you so much for your comments. We changed the control example in Figure 3J. For sciatic nerve regeneration experiments, the sciatic nerve was exposed at the sciatic notch by a small incision 2 days after the in vivo electroporation. The nerve was then crushed, and the crush site was marked with a 11-0 nylon epineural suture. After surgeries, the wound was closed, and the mice were allowed to recover. Three days after the sciatic nerve crush, the whole sciatic nerves from the perfused animals were dissected out and postfixed overnight in 4% PFA at 4°C. Before whole-mount flattening, it was confirmed that the place of epineural suture matched the injury site, and experiments were included in the analysis only when the crush site was clearly identifiable. Using whole mounted tissue, all identifiable EGFP-labeled axons in the sciatic nerve were manually traced from the crush site to the distal growth cone to measure the length of axon regeneration. (line 159-164).

(3) The authors then examined the role of the phosphatase MLCP in axon growth during regeneration. The authors first use a known MLCP blocker, phorbol 12,13-dibutyrate (PDBu), to show that is able to increase the levels of p-MLC, with a concomitant increase in the extent of axon regrowth of DRG neurons, both in permissive as well as non-permissive. The authors repeat the experiments using the knockdown of MYPT1, a key component of the MLC-phosphatase, and again can observe a growth-promoting effect (Figure 3).

The authors further show evidence for the growth-enhancing effect in vivo, in nerve crush experiments. The evidence in vivo deserves more evidence and experimental details (see comment 2). Some key weaknesses of the data were mentioned previously (unclear RNAi controls and duplication of shown images), but in this case, it is also not clear if there is a change only in the extent of growth, or also in the number of axons that are able to regenerate.

Thank you so much for your comments. We used same control as in vitro experiments (the pCMV-EGFP-N3 plasmid was from Clontech, Inc), and we also changed the control image in Figure 3J. For in vivo axon regeneration experiments, we measured the lengths of all identifiable EGFP-labelled axons in the sciatic nerve from the crush site to the distal axonal ends. The number of EGFP labeled regenerating axons were actually determined by the electroporation rate of EGFP, which is similar, but not identical, in different mice. Thus, our data only can show the differences in axon lengths among different experimental conditions. Such approach has been used in many of our previously published papers (e.g. Saijilafu et al. Nature Communications, 2011, Saijilafu et al. Nature Communications, 2013). (line 152-153).

(4) In the next set of experiments (presented in Figure 4) authors extend the previous observations in primary cultures from the CNS. For that, they use cortical and hippocampal cultures, and pharmacological and genetic loss-of-function using the above-mentioned strategies. The expected results were obtained in both CNS neurons: inhibition or knockdown of the kinase decreases axon growth, whereas inhibition or knockdown of the phosphatase increases growth. A main weakness in this set is that it is not indicated when (at what day in vitro, DIV) the treatments are performed. This is important to correctly interpret the results, since in the first days in vitro these neurons follow well-characterized stages of development, with characteristic cellular events with relevance to what is being evaluated. Importantly, this would be of value to understand whether the treatments affect axonal specification and/or axonal extension. Although these events are correlated, they imply a different set of molecular events.

The treatments were started from the initial of cell culture period, and this procedure may affect axon specification as the Reviewer point out. However, we mainly focused on axon length in our experiments, thus, for quantification of axon length, neurons with processes longer than twice the diameter of cell bodies were photographed, and the longest axon of each neuron was measured. We revised the manuscript as suggested by the reviewer (line 143-145).

The title of this section is misleading: line 241 "MLCK/MLCP activity regulated axon growth in the embryonic CNS"... the title (and the conclusion) implies that the experiments were performed in situ, looking at axons in the developing brain. The most accurate title and conclusion should mention that the evidence was collected in CNS primary cultures derived from embryos.

We have revised the manuscript as suggested by the reviewer (line 251).

(5) Performing nerve crush injury in CNS nerves (optic nerve and spinal cord), and the local application of PBDu, the author shows contrasting results (Figure 5). In the ON nerve, they can see axons extending beyond the lesion site due to PBDu. On the contrary, the authors fail to observe so in the corticospinal tract present in the spinal cord. The authors fail to discuss this matter in detail. Also, they accommodate the interpretation of the evidence in light of a process known as axon retraction, and its prevention by MLCP inhibition. Since the whole paper is on axon extension, and it is known that mechanistically axon retraction is not merely the opposite of axon extension, the claim needs far more evidence.

Thank you so much for your comments. Compared to optic nerve axons, corticospinal tract axons exhibit a reduced intrinsic axon growth capability. Consequently, we observed that PBDu stimulates optic nerve axon regeneration. However, unfortunately, we did not detect any enhancement in corticospinal tract axons beyond the injury site in SCI following the inhibition of myosin light chain phosphatase (MLCP) with PBDu.

In panel 5F and the supplementary data, the authors mention the occurrence of retraction bulbs, but the images are too small to support the claim, and it is not clear how these numbers were normalized to the number of axons labeled in each condition.

Thank you so much for your comments. In this study, we used a similar method from Ertürk et al. (2007) to quantify the retraction bulb. Both maximum width of the enlarged distal tip of the axon and the width of its immediately adjacent axon shaft was measured. Then, the ratio of these two widths was then calculated. An axonal tip was considered as a retraction bulb if its tip/shaft ratio exceeded 4. Averages number of retraction bulb were calculated from 3 sections in every mice for each group (n=5). (line 187-191).

[Ref] Ertürk A, Hellal F, Enes J, and Bradke F (2007). Disorganized microtubules underlie the formation of retraction bulbs and the failure of axonal regeneration. J. Neurosci 27, 9169–9180. [PubMed:17715353].

*(6) The author combines MLCK and MLCP inhibitors with Bleb, trying to verify if both pairs of inhibitors act on the same target/pathway (Figure 6). The rationale is wrong for at least two reasons.
a- Because both lines of evidence point to contrasting actions of NMII on axon growth, one approach could never "rescue" the other.*

If MLCK regulates axon growth through the activation of Myosin, the inhibitory effect of ML-7 (an MLCK inhibitor) on axon growth might be influenced by Bleb, a NMII inhibitor. However,

our findings reveal that the combination of Bleb and ML-7 does not alter the rate of axon outgrowth compared to ML-7 alone. This suggests that the roles of ML-7 and Bleb in axon growth are independent. It means MLCK may regulate axon growth independent of NMII activity.

b. Because the approaches target different steps on NMII activation, one could never "prevent" or rescue the other. For example, for Bleb to provide a phenotype, it should find any p-MLC, because it is only that form of MLC that is capable of inhibiting its ATPase site. In light of this, it is not surprising that Bleb is unable to exert any action in a situation where there is no p-MLC (ML-7, which by inhibiting the kinase drives the levels of p-MLC to zero, Figure 4A). Hence, the results are not possible to validate in the current general interpretation of the authors. (See 'major concern').

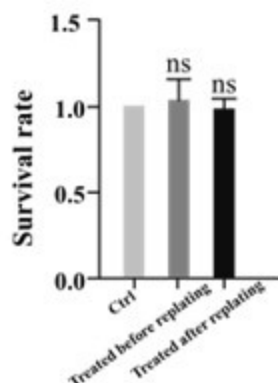
The reported mechanism of blebbistatin is not through competition with the ATP binding site of myosin. Instead, it selectively binds to the ATPase intermediate state associated with ADP and inorganic phosphate, which decelerates the phosphate release. Importantly, blebbistatin does not impede myosin's interaction with actin or the ATP-triggered disassociation of actomyosin. It rather inhibits the myosin head when it forms a product complex with a reduced affinity for actin. This indicates that blebbistatin functions by stabilizing a particular myosin intermediate state that is independent of the phosphorylation status of myosin light chain (MLC).

[Ref] Kovács M, Tóth J et al. Mechanism of blebbistatin inhibition of myosin II. J Biol Chem. 2004 Aug 20;279(34):35557-63. doi: 10.1074/jbc.M405319200.

(7) In Figure 7, the authors argue that the scheme of replating and using ML7 before or after replating is evidence for a local cytoskeletal action of the drug. However, an alternative simpler explanation is that the drug acts acutely on its target, and that, as such, does not "survive" the replating procedure. Hence, the conclusion raised by the evidence shown is not supported.

In our study, we meticulously assessed the neuronal survival rates across various experimental groups. The findings indicate no significant variation in survival rates among the groups. This suggests that the drug treatment exerts no discernible influence on cell viability but primarily modulates axonal elongation."

Author response image 1.



(8) In Figure 8, the authors show that the inhibitory treatments on MLCK and MLCP (ML7 and PRBu) alter the morphology of growth cones. However, it is not clear how this is

correlated with axon growth. The authors also mention in various parts of the text that a local change in the growth cone is evidence for a local action/activity of the drug or enzyme. However, the local change $\leftrightarrow$ local action is not a logical truth. It can well be that MLCK and MLCP activity trigger molecular events that ultimately have an effect elsewhere, and by looking at "elsewhere" one observes of course a local effect but is not because the direct action of MLCK or MLCP are localized. To prove true localized effects there are numerous efforts that can be made, starting from live imaging, fluorescent sensors, and compartmentalized cultures, just to mention a few.

About the relationship between growth cone size and its growth rate, the previous published literatures found that a fast-growing axon tended to have small growth cones (Mason C. et al. 1997). A recent study on *Aplysia* further supports this by noting that growth cones enlarge significantly when axonal elongation halts (Miller and Suter, 2018). Consistent with these findings, our data indicate that inhibiting MLCP with PDBu treatment leads to a reduction in growth cone size, which in turn promotes axon regeneration.

[Ref] Mason CA, Wang LC. Growth cone form is behavior-specific and, consequently, position-specific along the retinal axon pathway. *J Neurosci*. 1997; 13:1086–1100. [PubMed: 8994063]

[Ref] Miller KE, Suter DM. An Integrated Cytoskeletal Model of Neurite Outgrowth. *Front Cell Neurosci*. 2018 Nov 26;12:447. doi: 10.3389/fncel.2018.00447. eCollection 2018.

References:

- (1) Eun-Mi Hur 1, In Hong Yang, Deok-Ho Kim, Justin Byun, Saijilafu, Wen-Lin Xu, Philip R Nicovich, Raymond Cheong, Andre Levchenko, Nitish Thakor, Feng-Quan Zhou. 2011. Engineering neuronal growth cones to promote axon regeneration over inhibitory molecules. *Proc Natl Acad Sci U S A*. 2011 Mar 22;108(12):5057-62. doi: 10.1073/pnas.1011258108.
- (2) Garrido-Casado M, Asensio-Juárez G, Talayero VC, Vicente-Manzanares M. 2024. Engines of change: Nonmuscle myosin II in mechanobiology. *Curr Opin Cell Biol*. 2024 Apr;87:102344. doi: 10.1016/j.ceb.2024.102344.
- (3) Karen A Newell-Litwa 1, Rick Horwitz 2, Marcelo L Lamers. 2015. Non-muscle myosin II in disease: mechanisms and therapeutic opportunities. *Dis Model Mech*. 2015 Dec;8(12):1495-515. doi: 10.1242/dmm.022103.

Reviewer #2 (Public review):

Summary:

Saijilafu et al. demonstrate that MLCK/MLCP proteins promote axonal regeneration in both the central nervous system (CNS) and peripheral nervous system (PNS) using primary cultures of adult DRG neurons, hippocampal and cortical neurons, as well as in vivo experiments involving sciatic nerve injury, spinal cord injury, and optic nerve crush. The authors show that axon regrowth is possible across different contexts through genetic and pharmacological manipulation of these proteins. Additionally, they propose that MLCK/MLCP may regulate F-actin reorganization in the growth cone, which is significant as it suggests a novel strategy for promoting axonal regeneration.

Strengths:

This manuscript presents a comprehensive array of experimental models, addressing the biological question in a broad manner. Particularly noteworthy is the use of multiple in vivo models, which significantly strengthens the overall validity of the study.

We thank the Reviewer for taking time to review our manuscript, and we really appreciated the positive comments from the Reviewer.

Weaknesses:

The following aspects apply:

(1) The manuscript initially references prior research by the authors suggesting that NMII inhibition enhances axonal growth and that MLCK activates NMII. However, the study introduces a contradiction by demonstrating that MLCK inhibition (via ML-7 or siMLCK) inhibits axonal growth. This inconsistency is not adequately addressed or discussed in the manuscript.

Thank you for reviewer's very good comments. As suggested by Reviewer, we discuss more detail it in our revised manuscripts (line 357-368; line373-374).

(2) While the study proposes that MLCK/MLCP regulates F-actin redistribution in the growth cone, the mechanism is not explored in depth. The only figure showing how pharmacological manipulation affects the growth cone suggests that not only F-actin but also the microtubule cytoskeleton might be affected, indicating that the mechanism may not be specific. A deeper exploration of this relationship in DRG neurons, in addition to cortical neurons, as shown in the study, would be beneficial.

Thank you for your insightful suggestion. However, our study primarily focuses on actin and myosin dynamics in the context of axonal elongation, as indicated by our direct observations in growing dorsal root ganglia (DRGs). Athamneh et al. (2017) elegantly demonstrated that the bulk movement of microtubules (MTs), rather than their assembly, predominantly drives MT advance during axonal elongation. Consequently, our manuscript concentrates on the actomyosin system, which is central to our findings. While the role of MTs in axonal growth is indeed significant and fascinating, the data we present is predominantly concerned with the actomyosin mechanism.

[Ref] Athamneh, A. I. M. et al. Neurite elongation is highly correlated with bulk forward translocation of microtubules. Scientific Reports 7, (2017).

(3) In the sciatic nerve injury experiments, it would be crucial to include additional controls that clearly demonstrate that siMYPT1 treatment increases MLCP in the L4-L5 ganglia. Additionally, although the manuscript mentions quantifying axons expressing EGFP, the Materials and Methods section only discusses siMYPT1 electroporation, which could lead to confusion.

Thank you for your suggestion. However, due to the unavailability of a suitable commercial MLCP antibody, we were unable to directly detect MLCP expression. Instead, we assessed the phosphorylation level of myosin light chain (MLC) as a proxy to indicate that siMYPT1 transfection effectively downregulates MLCP activity in L4/5 dorsal root ganglia (DRG). This approach was taken to ensure the integrity of our findings despite the limitations in antibody availability.

About the electroporation method section, we have now included detailed information about the control plasmid used in our experiments to ensure a clear understanding of our experimental setup and to validate our results. A 1 µl solution containing indicated siRNAs together with the plasmid encoding EGFP (pCMV-EGFP-N3) was then microinjected into the L4-L5 DRG..... (line 152-153).

(4) In some panels, it is difficult to differentiate the somas from the background (Figures 3, 4, 7). In conditions where images with shorter axonal lengths are represented, it is unclear whether this is due to fewer cells or reduced axonal growth (Figures 2, 4, 6).

In the original submission, there was some loss of image quality while converting the TIFF to PDF. We improved the quality of images in our revised manuscripts.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

There are a number of typos and language errors that should be thoroughly revised. For example, line 219: "It is well known that the opposite role of MLCK and MLCP to regulate the MLC phosphorylation status". The term "opposite role" is vague. Using "opposite roles" and specifying that they are in regulating MLC phosphorylation status clarifies the relationship between MLCK and MLCP. Also, the original phrase "to regulate" was not correctly integrated into the sentence. Rephrasing it to "in regulating" makes the role of MLCK and MLCP clearer.

We have revised the manuscript as suggested by the reviewer (line 229).

In the same line, there is a high number of panels that are not referred to in the text or references for panels that have another letter. Just to mention a few:

- line 199: "(Figure 1F, G)", → BUT figure 1 contains no G panel.

We have revised the manuscript as suggested by the reviewer (line 209).

- line 203: "The results showed that ML-7 administration led to a significant reduction in MLC phosphorylation levels (Figure 2A, B) and impaired axonal growth in sensory neurons (Figure 2C, D). → BUT panel C is related to A and B, and only D and E show impaired axonal growth.

We have revised the manuscript as suggested by the reviewer (line 214; line 215; line 217; line 219).

Reviewer #2 (Recommendations for the authors):

(1) Improving the quality of the images would significantly strengthen the results presented.

In the original submission, there was some loss of image quality while converting the TIFF to PDF. We improved the quality of images in our revised manuscripts.

(2) The representative images of controls do not always show the same number of cells or axonal growth (e.g., Figure 4).

We have changed some images as suggested by the reviewer.

(3) The text has citation errors when referring to the figure labels.

Upon thorough review, we have carefully examined our manuscript and have made the necessary corrections to address the identified errors. We appreciate the opportunity to

enhance the quality of our work and believe that these revisions have significantly improved the clarity of our manuscript.

| (4) *What happens to MLCK levels when MLCP activity is inhibited in the optic nerve?*

Upon analyzing our experimental data, we observed no significant alterations in the protein levels of MLCK when the activity of MLCP was inhibited. This finding suggests that the regulatory mechanisms governing MLCK expression may not be directly influenced by short-term MLCP inhibition. It is plausible that the duration of the inhibition period was insufficient to elicit a detectable change in MLCK expression levels.

| (5) *The text in line 266: "In contrast, local PBS administration at the injury site or intravitreal PDBu injection induced little axon regeneration beyond the injury site (Figure 5 A-C)." However, this is not reflected in the figure.*

In our revised manuscript, we have provided a more precise description of our findings: In contrast, local PBS administration at the injury site or intravitreal PDBu injection did not significantly enhance axon regeneration beyond the injury site (Figure 5 A-C). This observation suggests that the only treatment employed in the injury site (the inhibition of MLCP activity within the growth cone) effectively promote axonal growth. (line 276-279).

| (6) *Line 287: The phrase "Consistent with our previous study" requires a citation to support it.*

We added the reference paper; Consistent with our previous study 1, the inhibition of myosin II activity with 25 μ M blebbistatin markedly promoted axonal growth (Figure 6A, B). (line 298)

| (7) *Line 333: The paper cited by Yu P et al. (2012) does not mention MLCK or p-MLC, so it appears to be misquoted.*

Thank you for comments. We rechecked this cited paper and confirmed that the author provided the western data C in the supplementary figure 1, it showed that Bleb did not alter the phosphorylation status of MLC.

<https://doi.org/10.7554/eLife.101250.2.sa0>